



Milos Restaurant Roof, Atlantis The Royal, Palm Jumeirah, Dubai

Engineering Feasibility Report



**September 2023
Rev. 01**

CONTENTS		
1.0	Introduction	
	1.1	Purpose of the Report
	1.2	Approach
	1.3	Building History
	1.4	Scope and Limitations of the Report
2.0	Executive Summary	
3.0	General Building Description	
4.0	MEP Feasibility Study	
5.0	VT Feasibility Study	
6.0	FLS Feasibility Study	
7.0	Structural Feasibility Study	

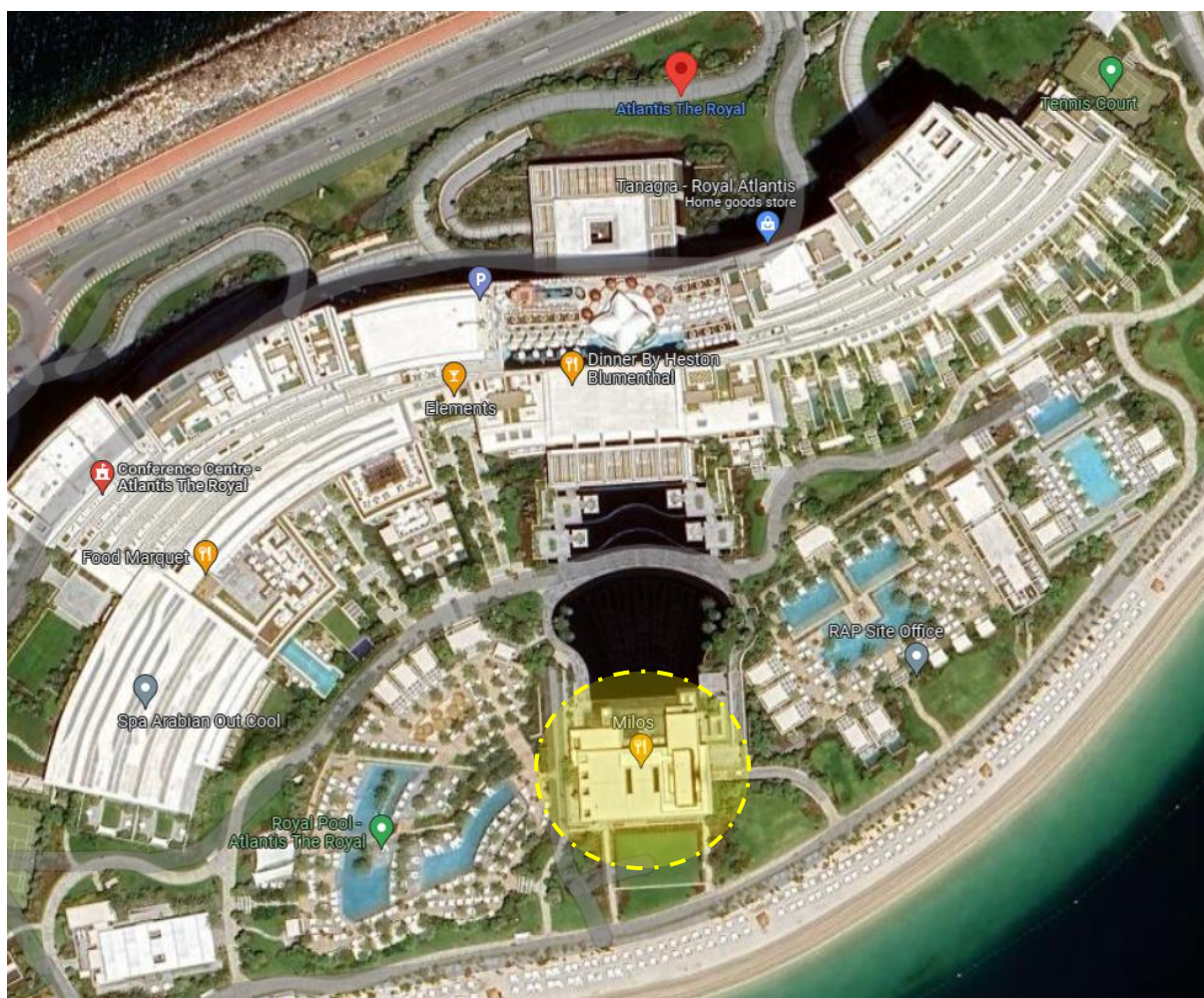
	Appendices
	Appendix A – Structural As-Built & Shop Drawings

1.0 INTRODUCTION

1.1 Purpose of the Report

Astbury Shepherd Gilmour (ASG) Engineering Consultants was commissioned by The Royal Atlantis Hotel, Dubai, Palm Jumeirah to undertake feasibility of using the existing roof top as a roof bar at Milos Restaurant.

Milos rooftop bar (as marked by yellow on the figure below) can accommodate up to 200 guests with live entertainment, DJs, and a perfect venue for special events.



Milos Restaurant Location

This report was prepared for the benefit of the building owner/occupier.

1.2 Approach

In preparation of the report, a site visit has been carried out by ASG which is based on the following approach:

- Site Inspection
Initial site visit on 7th August 2023; a general check with an overall impression of the building structure had been done to explore the proposed modifications.
- Review of As-Built, and Shop drawings
In advance of a visit to the site, ASG was provided with the As-Built & shop drawings as included in Appendix A. The review has not included a full review of the design content of the drawings.
- Review Analysis and Report Preparation
Following the inspection and drawing review, the information from both has been collected together and used in the preparation of this report.

1.3 Building History

The building was constructed 1 year ago as an in-situ concrete structure and concrete deck slab on steel truss for the roof at Milos restaurant. The building is designed by (IBI Group Architects & Dynamic Engineering Consultants).

1.4 Scope and Limitations of the Report

This report is based on the instructions of The Royal Atlantis Hotel, Dubai, Palm Jumeirah.

This is not a full design check of the existing structure, MEP and FLS Systems

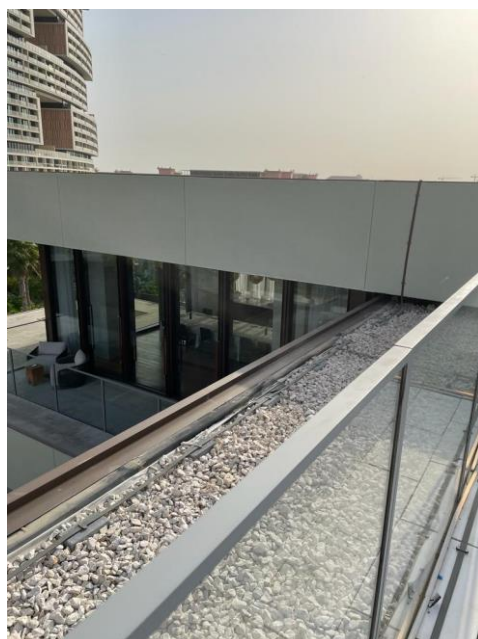
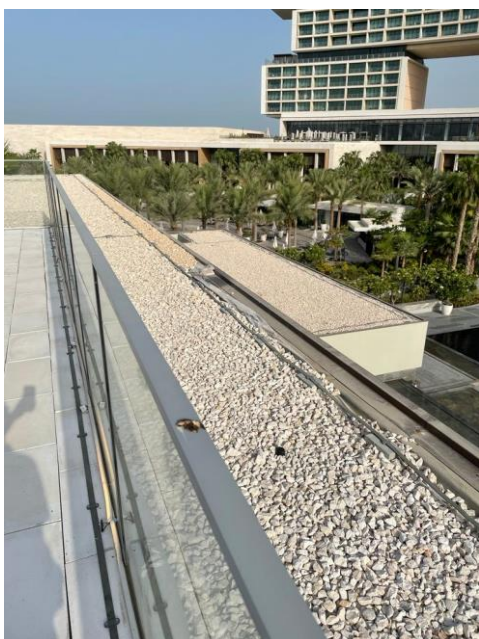
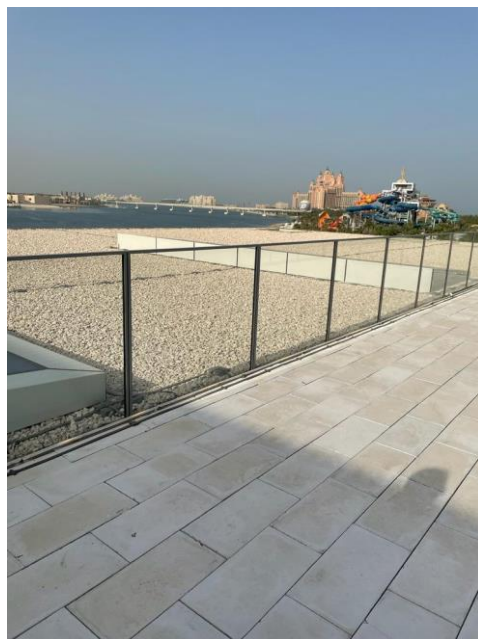
The full scope of the report is to:

- Carry out a preliminary Structural, MEP and FLS study of the impact of the proposed modifications on the building and report on the findings, taking into account any information provided.
- The report is based on the information provided as listed in the appendix and the visual site visit made on 7th August 2023.
- No opening up works were carried out to investigate the existing building.
- The report is a professional opinion of the structure, MEP and FLS Systems
- The report is for The Royal Atlantis Hotel, Dubai, Palm Jumeirah and ASG offer no duty of care or liability to any other parties.
- No remedial work details are provided in this report as it is a report to understand the feasibility of the proposed modifications.

2.0 EXECUTIVE SUMMARY

The Royal Atlantis Hotel, Dubai, Palm Jumeirah are planning for modifications to the new hotel as noted below.

The existing standalone Milos Greek Restaurant is required to have the current rooftop converted into a roof top bar/lounge with shade structures above. We understand the external deck will not require HVAC, nor will it have a functional kitchen – though there will be an allowance for a small bar of some description complete with wet services (water supply & drainage).



Milos Restaurant – Roof Floor

These proposed modifications have been considered for impact as follows:

2.1 MEP

This report specifically studies the required MEP (Mechanical, Electrical & Public Health) services to facilitate a new Milos rooftop bar / lounge located above the existing indoor Milos Restaurant.

To carry out this report we have assumed that the full roof area will be utilized (excluding roof lights) and the expected occupancy is 200 guests. The area is fully outside and therefore no permanent HVAC solutions are to be proposed however we have assumed the use of portable type evaporative coolers when required during summer.

The key elements which will be focused on to determine the feasibility of the roof terrace proposal are mainly relating to the power required in addition to some considerations regarding a water supply & drainage to serve a small bar facility. It is however noted that any roof penetrations are to be avoided if possible.

2.2 VT

This initial report outlines the information that has been used for the Vertical Transportation evaluation for this project as provided September 2023

This analysis has been conducted using defined and 'assumed' population levels.

Elevate Ver 9 has been used for this Traffic Analysis (Elevate is an Industry standard and unbiased traffic analysis programme accepted around the world as being the most accurate system).

2.3 FLS

This section of the report presents the fire and life safety feasibility study for the Milos Greek Restaurant, located at the Royal Atlantis, Palm Jumeirah.

The restaurant is a stand-alone structure which consists of:

- Ground floor – Dining and kitchen services
- First Floor – Private dining
- Roof – storage and MEP

The proposal is to utilize the rooftop for outdoor dining.

This feasibility study will provide the requirements as stated within the UAE fire and life safety code 2018 edition, taking into consideration the maximum number of persons permitted to be on the roof based on the available exits.

The FLS section of the report will address the following:

- Means of egress analysis
- Fire rating of walls and doors
- Fire protection systems
- Fire detection and alarm system requirements

It should be noted that the Dubai Civil Defence Approved Drawings have not been provided, therefore the assumptions made shall be revalidated based on the approved Dubai Civil Defence drawings.

2.4 Structural

ASG evaluated the structural capacity of existing Roof Top deck slab, steel trusses & beams in the standalone Milos Greek restaurant and validated it against the new live load, and additional planter boxes at two sides of the restaurant for the roof top Bar /Lounge.

ASG found that the existing roof structure cannot support the loads of the new roof live loads without strengthening of the existing steel roof trusses and adding additional structural steel members to enable the existing roof slab to carry out the new loads. These strengthening works can only be carried out from below.

ASG has studied an alternative option to add a new independent roof deck above the existing roof in order to support the new loads. This would require the new roof to transfer the new loads through to the existing columns below and due to the large spans would need to be based around a structural truss system (similar to the existing structure below).

3.0 GENERAL BUILDING DESCRIPTION

The building was constructed 1 year ago as an in-situ concrete structure and concrete deck slab on steel truss for the roof at Milos restaurant. The building is designed by (IBI Group Architects & Dynamic Engineering Consultants).

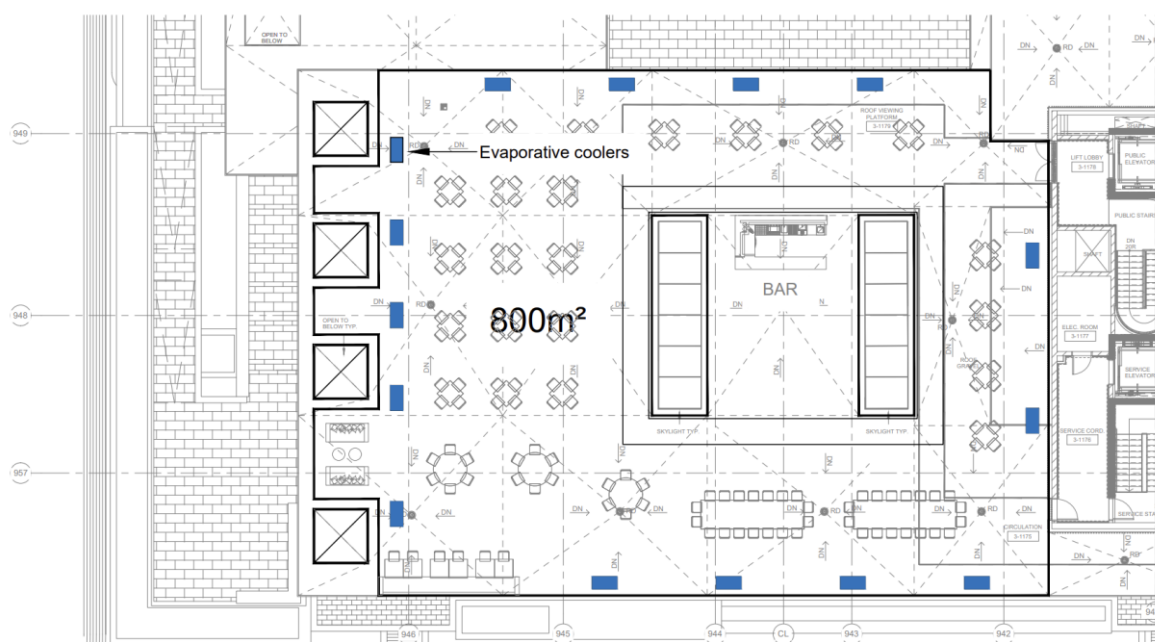
The structure for standalone Milos Greek restaurant consists of a concrete deck slab at level +19.24 supported by 3 steel trusses of maximum span of 21.30 m and intermediate secondary steel beams at variable spacing. The main trusses are rested over 9 concrete columns of 8.50 m height.

4.0 MEP FEASIBILITY STUDY

4.1 Mechanical

4.1.1 HVAC Services

As mentioned in the introduction, there will be no fixed / permanent HVAC solution imposing additional loads on the buildings HVAC network as the roof terrace is outdoors. We have however considered some portable evaporative coolers which can be used in the hotter months as per below:



The electrical load required is minimal, 240W per unit (to be reviewed in the electrical section of this report). The area covered by each cooler (shown in blue around the roof perimeter) ranges from 25-50m² depending on the manufacturer / model. Something to note is the water consumption of the units and their storage capacity of 55L. We believe a Bop tap should be installed somewhere on the roof to enable easy refilling of the units.

Upon carrying out a site inspection, it was noted that there was condensation forming on the ceiling diffuser within the lobby areas and staircase, below the rooftop access. We believe this is due to the excessive opening of the roof exit door. Our first recommendation would be to ensure the door ironmongery is working correctly to swing shut, and the seal installed correctly and not corroded. Secondly, we would suggest rebalancing the ventilation system to increase the positive pressure to the area to ensure the space adjacent the exit door is positively pressurized.

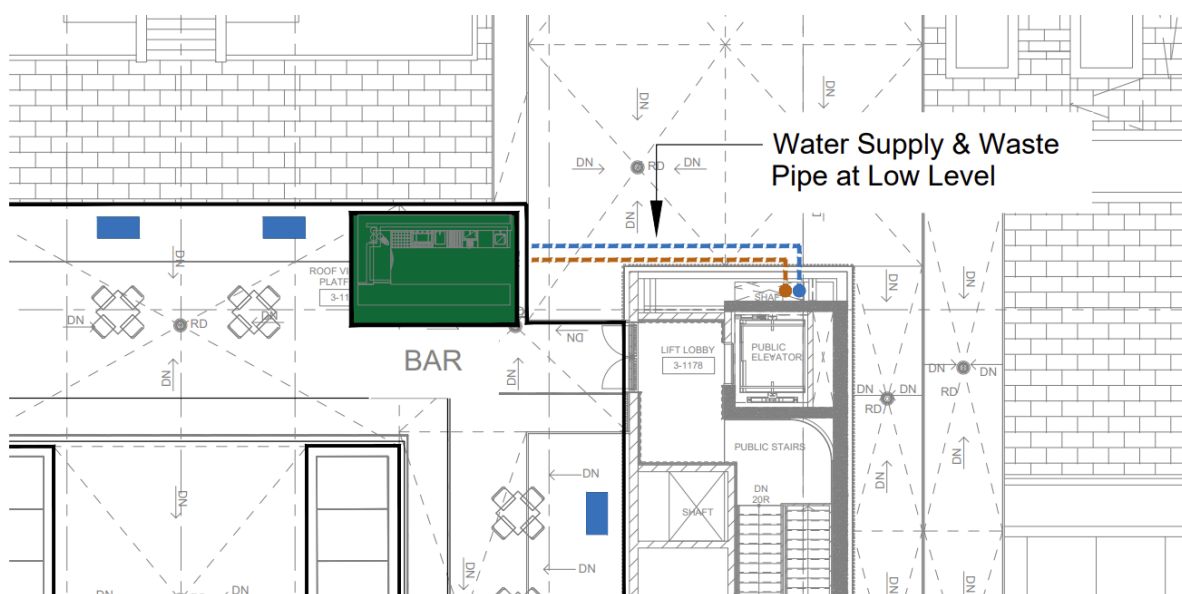
Lastly, if the above recommendations do not solve the condensation problem experienced on the air diffusers, we believe an air curtain should be installed to help prevent ingress of hot humid air which is condensing on the diffuser. The air curtain should be an ambient type given that there is only a 20mm chilled water pipe within the roof level lobby which cannot accommodate any additional significant chilled water flowrate.

4.2 Public Health

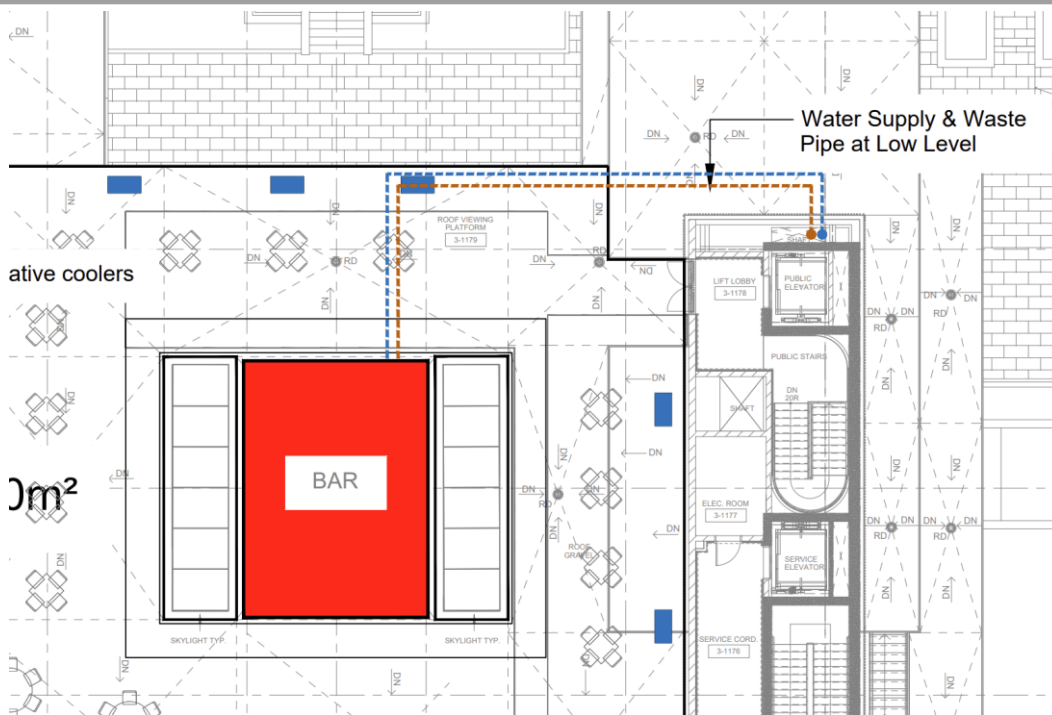
4.2.1 Water Supply & Drainage

If it is desired to have a functional bar inclusive of wet services to supply a sink, we believe the bar should be located as close to the stair core as possible. Within this riser, there will be wet services / waste serving a restroom on the below floor.

Considering the below sketch, we have indicatively positioned the bar approximately 8m away from the riser. This means a 50mm waste pipe with a fall of 1 in 100 will require a clearance above the floor of approximately 150mm. The water supply could run alongside this, and so a stepover or floor buildup would be required to accommodate these services. A bib tap could also be provided from this point however the pipe size needs to be studied upon receipt of the water supply drawings.

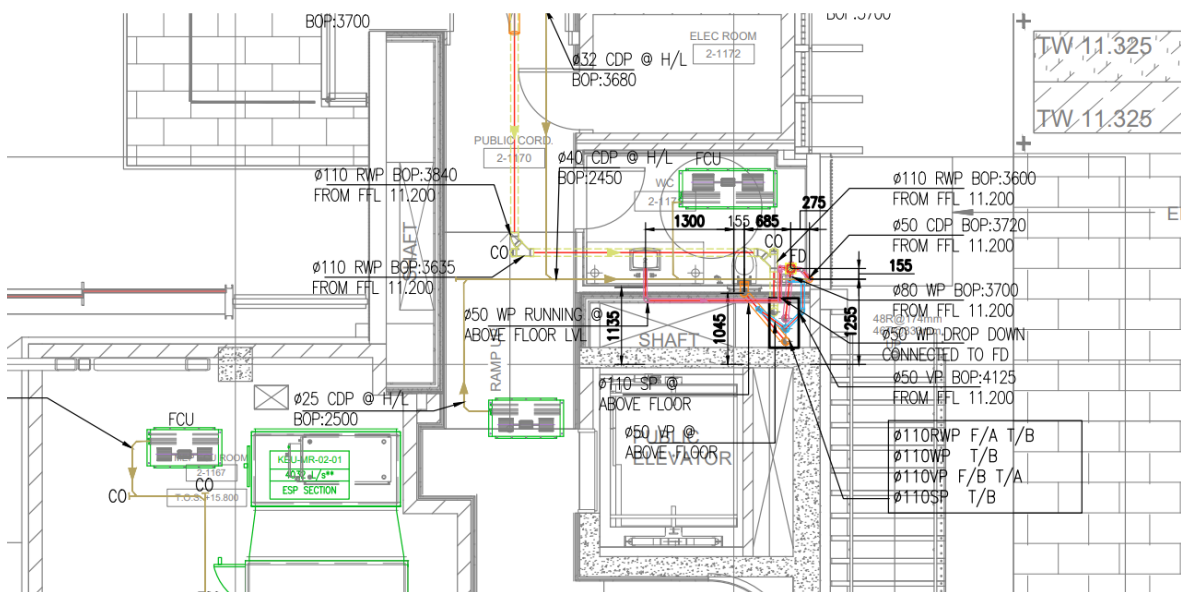


If it is preferred to have a centrally positioned bar with wet services utilities (possible bar area shown below in red) the floor depth / buildup or stepover required would be circa 350mm due to a waste pipe length of 25m, hence why it is better to locate it as per above shaded green area. The other option would be to core through the roof to the ceiling void below however we are aware this is not preferred.

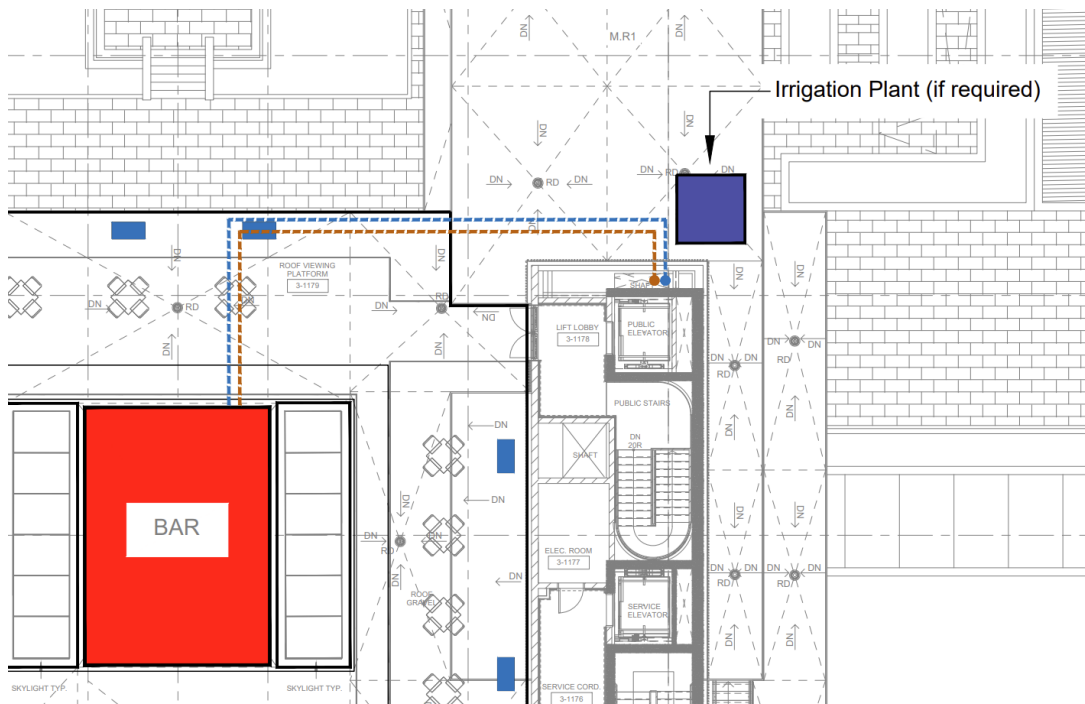


Rather than raising the entire roof floor by the aforementioned 350mm it may be possible to have a step up isolated around where the services are located. This needs to be considered architecturally in line with any other raised floor requirements, to facilitate rainwater channels / electrical floor boxes for example.

A water supply drawing has not been provided however we can see the below restroom here, and the waste pipe serving it. As mentioned, we assume there will also be a water supply here which could be extended.



Finally, landscaping has been excluded from this study however it's worth mentioning that irrigation can be catered for by manual operation via supply of water from a newly installed bib tap. Alternatively, depending on client requirements, a small automatic system complete with pump, tank and controls could be proposed on an area of the roof not accessible by guests (indicatively shown below). Pipes would then be distributed to the plants via pipework within the floor buildup / void.



4.2.2 Rainwater Outlets

To ensure no additional coring is considered for the rooftop slab, and maintaining the current warranty on the waterproofing we recommend keeping the existing locations of the roof rainwater outlets. As such, the roof drainage strategy we propose would be as follows:

- Ensure a channel drain is considered in the architectural floor buildup of what we assume would be some type of paving system.
- The channel drain should align with the current rainwater outlet grid formation.
- The discharge point within the channel drain trench would then be plumbed directly into the existing rainwater outlet to ensure the current roof drainage system can maintain the level of hardscape and surface run originally designed.

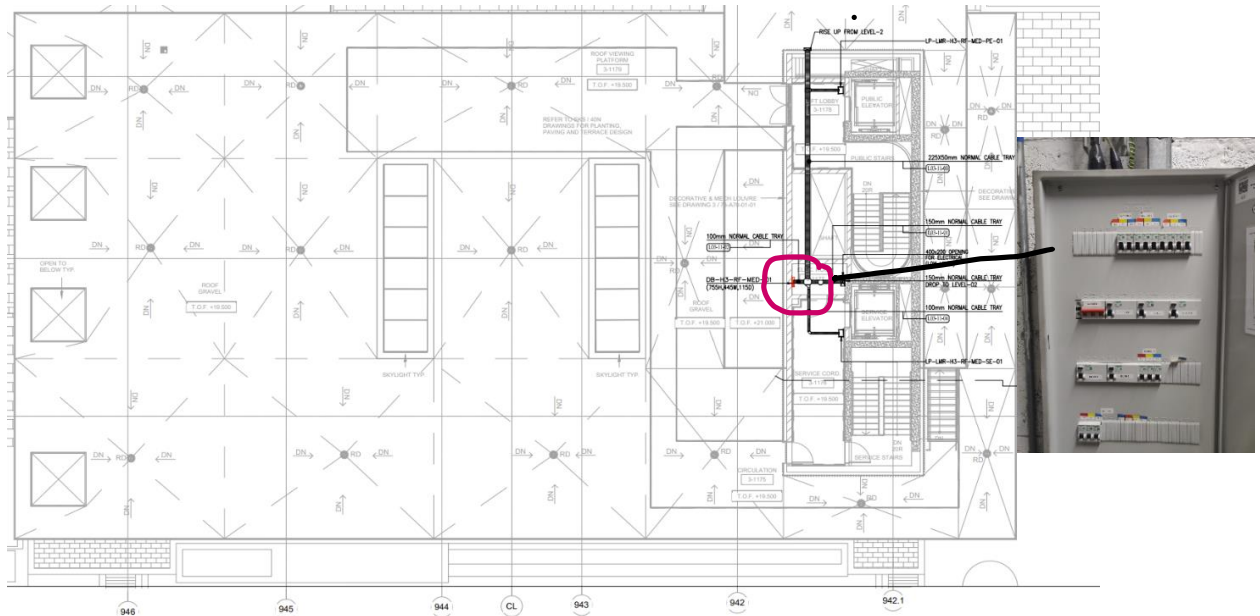
Mechanical / PH Conclusion:

As per initial studies, HVAC, drainage and water supply can be accommodated to supply a small bar and evaporative coolers from a HVAC perspective. Finer details such as the floor buildup depth will need to be investigated, also the irrigation requirements need to be confirmed by the client.

4.3 Electrical System – Power

4.3.1 Existing Load

As per below As-built Drawing (TRA-SBJ-ABD-EL-L03-00011-00) received on 2023-23-08, the imagery shows a dedicated DB-H3-RF-MED-01 in roof Electrical room that is fed from Ground Floor SMDB-H3-GF-MED-01. Existing roof DB has an incoming 40A TP isolator with a total connected load of 4.4W kilowatts. Incoming cable size, 1x4C 10mm² Cu/XLPE/SWA/LSF + 1x1C 10mm² Cu/G/Y/LSF ECC.



DB-H3-RF-MED-01 – Load Schedule (Existing)

PROJECT:		THE ROYAL ATLANTIS RESORT AND RESIDENCES																				LOCATION OF ELECTRICAL ROOM (2-1172)			
DB REF:		DB-H3-RF-MED-01																				DATE:			
FED FROM:		SMDB-H3-GF-MED-01																				INCOMING CABLE SIZE: 1x4C 10mm2 Cu/XLPE/SWA/LSF + 1x1C Cu/G/Y/LSF ECC			
INCOMER:		40A TP ISOLATOR																							
Rating Of INCOMER	RATING OF ELCB	SL No.	CCT No.	MCB Rtg. In A	CCT. Cable Size mm²	ECC Cable Size mm²	Room / Area	CONNECTED LOADS/POINTS												LOAD PER CIRCUIT - W			Diversity Factors	Circuit load after diversity (W)	REMARKS
								LIGHTING	NEON LTG	EX. FAN	Curt. Mot.	13A S/O	13A T/O	13A SPUR	FCU	H/D	20A DP SW	ISO	MISC	Watts/ Unit	R	Y			
40A TP-ISOLATOR	40A 4P 100mA ELCB	01	R1	16	2.5	2.5	SERV CORR 3-1176	4											10	40			0.9	36.0	M-BD1
		02	Y1	16	2.5	2.5	PUBLIC STAIRS	26.4											4.6		121.44		0.9	109.3	ML3x24mos
		03	B1	16	2.5	2.5	LIFT LOBBY 3-1778	4											10			40	0.9	36.0	MD9x2"2
		04	R2	16	2.5	2.5	ELEC ROOM 3-1777	1											58	58			0.9	52.2	B1 TYPE LIGHT
		05	Y2	16	2.5	2.5	PUBLIC STAIRS	11.9											10			119	0.9	107.1	ML1 TYPE LIGHT
		06	B2	16	2.5	2.5	FACADE LIGHTING -LEVEL 3	75											10.29			771.75	0.9	694.6	MDR 1.01
		07	R3				SPACE																		
		08	Y3				SPACE																		
		09	B3				SPACE																		
		10	R4				SPACE																		
		11	Y4				SPACE																		
		12	B4				SPACE																		
	40A 4P 100mA ELCB	13	R5	20	4.0	4.0	LIFT LOBBY 3-1778												213	213			0.9	191.7	FCU-MR-03-03
		14	Y5	20	4.0	4.0	SERV CORR 3-1176												157		157		0.9	141.3	FCU-MR-03-01
		15	B5	20	4.0	4.0	SERV CORR 3-1176												123		123	0.9	110.7	FCU-MR-03-02	
		16	R6				SPACE																		
		17	Y6				SPACE																		
		18	B6				SPACE																		
	40A 4P 30mA ELCB	19	R7	20	4.0	4.0	LIFT LOBBY 3-1778						2						200	400			0.9	360.0	
		20	Y7	20	4.0	4.0	ELEC/SERVICE RM/STAIRS						2	1					200/400		800		0.9	720.0	
		21	B7	20	4.0	4.0	ELEC ROOM 3-1777								1				400			400	0.9	360.0	POWER TO LCP
		22	R8	20	4.0	4.0	ROOF VIEWING PLATFORM (3-1179)								2				400	800			0.9	720.0	RF1 14473
	40A 4P 30mA ELCB	23	Y8	20	4.0	4.0	SERV CORR 3-1176								1				400		400		0.9	360.0	LIFT CONTROL FOR MS-01& MP-01
		24	B8				SPACE																		
TOTAL LOAD PER PHASE																		1511	1597	1335		3999			
TOTAL CONNECTED LOAD (TCL in kW):																		4.44							

4.3.2 Proposed 1

Power Load Calculation with Indicative Loads (Assume)

SN	EQUIPMENT LIST	QTY	POWER CONSUMPTION (Watts)	ELECTRICAL LOAD (W)
1	EVAP COOLERS	15	240.00	3600
2	DJ BOOTH - AV	2	1000.00	2000
3	REFRIGERATED DISPLAY	1	1500.00	1500
4	LIGHTING	5	300.00	1500
Total Connected Load(W)				8600.00
Total Max. Demand Load(W)				7740.00
Total Connected Load(kW)				8.6

DB-H3-RF-MED-01 – Load Schedule (with Indicative Load)

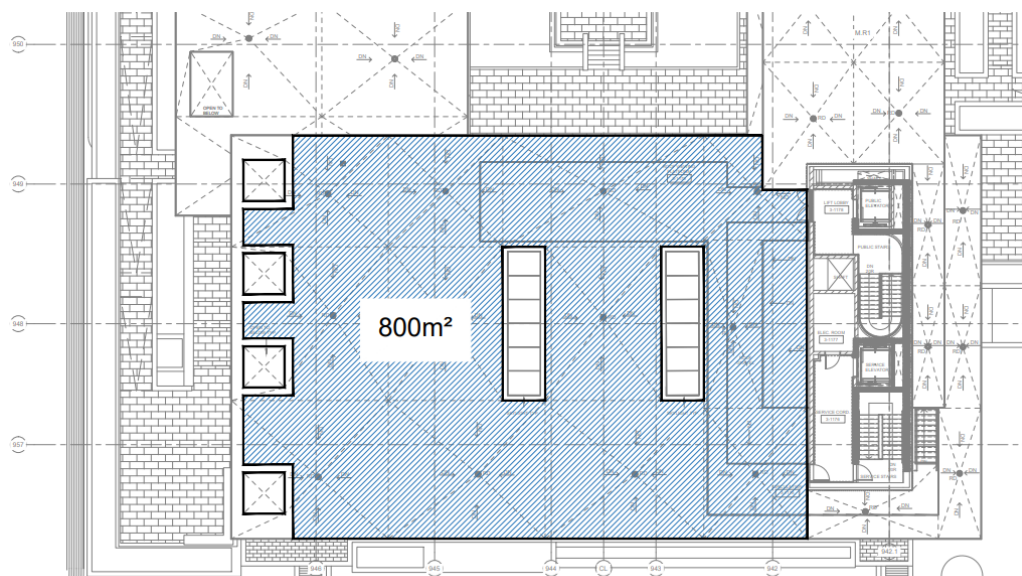
PROJECT:		THE ROYAL ATLANTIS RESORT AND RESIDENCES										LOAD DISTRIBUTION SCHEDULE										LOCATION OF ELECTRICAL ROOM (2-1172)				
DB REF:		DB-H3-RF-MED-01																				DATE:				
FED FROM:		SMDB-H3-GF-MED-01																				INCOMING CABLE SIZE: 1x4C 10mm2 Cu/XLPE/SW/LSF + 1x1C Cu/G/Y/LSF ECC				
INCOMER:		40A TP ISOLATOR																								
Rating Of INCOMER	RATING OF ELCB	SL No.	CCT No.	MCB Rtg. In A	CCT Cable Size mm²	ECC Cable Size mm²	Room / Area	CONNECTED LOADS/POINTS												Watts/ Unit	LOAD PER CIRCUIT - W			Diversity Factors	Circuit load after diversity (W)	REMARKS
								LIGHTING	NEON LTG	EX. FAN	Curt Mtr.	13A S/O	13A T/O	13A S/PUR	FCU	H/D	20A DP SW	IND SOCKET	MISC		R	Y	B			
40A TP ISOLATOR	40A 4P 100mA ELCB	01	R1	16	2.5	2.5	SERV CORR 3-1176	4											10	40			0.9	36.0	M-BD1	
		02	Y1	16	2.5	2.5	PUBLIC STAIRS	26.4											4.6		121.44		0.9	109.3	ML3x24nos	
		03	B1	16	2.5	2.5	LIFT LOBBY 3-1778	4											10			40	0.9	36.0	MD9x2*2	
		04	R2	16	2.5	2.5	ELEC ROOM 3-1777	1											58	58			0.9	52.2	B1 TYPE LIGHT	
		05	Y2	16	2.5	2.5	PUBLIC STAIRS	11.9											10		119		0.9	107.1	ML1 TYPE LIGHT	
		06	B2	16	2.5	2.5	FACADE LIGHTING - LEVEL 3	75											10.29			771.75	0.9	694.6	MDR 1.01	
	40A 4P 100mA ELCB	07	R3	16	2.5	2.5	ROOF LIGHTING CIRCUIT 1	30											15	300			0.9	270.0	LED LIGHTS	
		08	Y3	16	2.5	2.5	ROOF LIGHTING CIRCUIT 2	30											15		300		0.9	270.0	LED LIGHTS	
		09	B3	16	2.5	2.5	ROOF LIGHTING CIRCUIT 3	30											15			300	0.9	270.0	LED LIGHTS	
		10	R4	16	2.5	2.5	ROOF LIGHTING CIRCUIT 4	30											15	300			0.9	270.0	LED LIGHTS	
		11	Y4	16	2.5	2.5	ROOF LIGHTING CIRCUIT 5	30											15		300		0.9	270.0	LED LIGHTS	
		12	B4	16	2.5	2.5	SPACE																			
	40A 4P 100mA ELCB	13	R5	20	4.0	4.0	LIFT LOBBY 3-1778									1			213	213			0.9	191.7	FCU-MR-03-03	
		14	Y5	20	4.0	4.0	SERV CORR 3-1176									1			157		157		0.9	141.3	FCU-MR-03-01	
		15	B5	20	4.0	4.0	SERV CORR 3-1176									1			123			123	0.9	110.7	FCU-MR-03-02	
	40A 4P 30mA ELCB	16	R6	20	4.0	4.0	ROOF BAR AREA						7						240	1680			0.9	1512.0	EVAP COOLERS	
		17	Y6	20	4.0	4.0	ROOF BAR AREA						8						240		1920		0.9	1728.0	EVAP COOLERS	
		18	B6	20	4.0	4.0	ROOF BAR AREA						1						1500			1500	0.9	1350.0	REFRIGERATED (7000 AV)	
	40A 4P 30mA ELCB	19	R7	20	4.0	4.0	LIFT LOBBY 3-1778						2						200	400			0.9	360.0		
		20	Y7	20	4.0	4.0	ELEC/SERVICE RM/STAIRS						2	1					200/400		800		0.9	720.0		
		21	B7	20	4.0	4.0	ELEC ROOM 3-1777								1				400			400	0.9	360.0	POWER TO LCP	
	40A 4P 30mA ELCB	22	R8	20	4.0	4.0	ROOF VIEWING PLATFORM (3-1179)								2				400	800			0.9	720.0	RF1 14473	
		23	Y8	20	4.0	4.0	SERV CORR 3-1176								1				400		400		0.9	360.0	LIFT CONTROL FOR MS-01& MP-01	
		24	B8	20	4.0	4.0	DJ BOOTH											2	1000			2000	0.9	1800.0	AUDIO EQUIPMENT	
TOTAL LOAD PER PHASE																			3791	4117	5135					
TOTAL CONNECTED LOAD (TCL in kW):																		11.04								

Reference to the above assume Indicative loads for Roof Bar & seating areas, the calculated power load demand is around 8.6 kilowatts. The existing roof DB is sufficient for this assume load, subject to the verification of the existing Ground Floor SMDB-H3-GF-MED-01 which the DB is fed from.

4.3.3 Proposed 2

Power Load Calculation based on Floor Area & Power Density

Below hatched area of 800m² of the main roof is assume as reference in deriving a feasible power demand for the proposed bar and roof seating areas. Take note of the possibility of a reduction in the electrical loads that is subject to the final architectural / ID layout at next design stage.



The following initial power load calculation has been conducted based on previous projects in the region, combined with BSRIA Rule of Thumb loads.

Isolator Sizing Rqmt: = Based on 800 m² floor area & 15 W/m² power demand factor based on open roof area.

$$= 800 \text{ sq. meter} \times 15 \text{ Watts per sq. meter} / 1.732 \times 400 \times 0.8$$

$$= \mathbf{21.65 \text{ Amps (12kW)}}$$

Existing Roof DB Isolator Size: = **40.0 Amps (17kW)**

The existing Isolator size allocated of 40A TP+N normal power is sufficient based on the above initial calculation.

Conclusion

As per the above Existing As-built drawing and schedule, roof DB has a possibility to accommodate an additional load of approximately 12.00kW which may be enough for the bar power requirements subject to existing MDB Schematic verification.

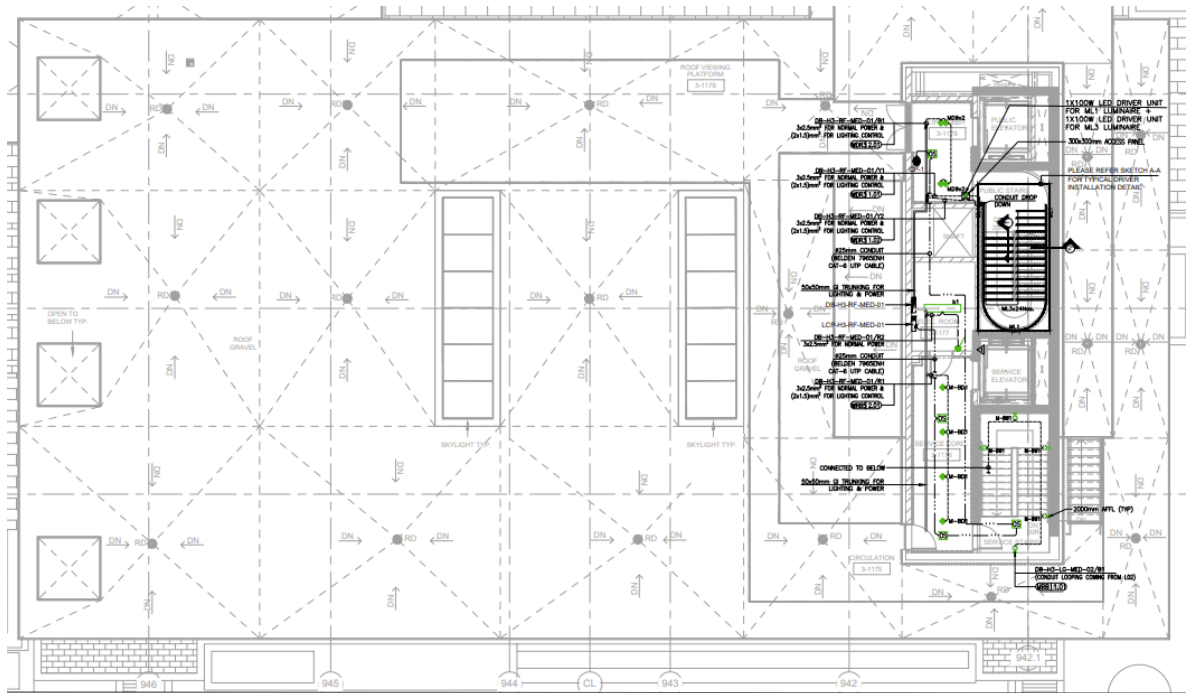
And in reference to proposal with assume indicative load and power density calculations, the existing roof DB is sufficient for this assume load as the existing Isolator size and incomer cable has been design to 40A TPN and still subject to the MDB capacity and spare loads. It is worth to highlight that the existing 40A TPN Isolator or DB can accommodate of maximum 17kW electrical load subject to voltage drop calculation.

Note: In case the load requirement exceeds 17 kilowatts, it is recommended to upgrade the cable size and circuit breaker size of the existing roof DB. Moreover, the upgrade may impact the existing upstream circuits distribution such as SMDB & MDB.

4.4 Lighting System

(By Specialist or the Interior Designer)

As per Drawing (TRA-SBJ-ABD-LT-L03-00311-00) received on 2023-23-08, roof shows a dedicated LCP-H3-RF-MED-01 inside the Electrical room with 6 connected lighting circuits. The existing ballast controller Philips Dynalite DDBC300-DALI for lighting control has 3 DALI Outputs and the dimmer controller (DDBC1200) features 12 independent outputs. The existing Lighting Control panel can still accommodate an additional ballast controller subject to the final lighting circuit requirement for roof Bar & Seating areas. It is crucial that the lighting specialist considers the existing controller capacity in their design. New lighting control panels can be installed as required.



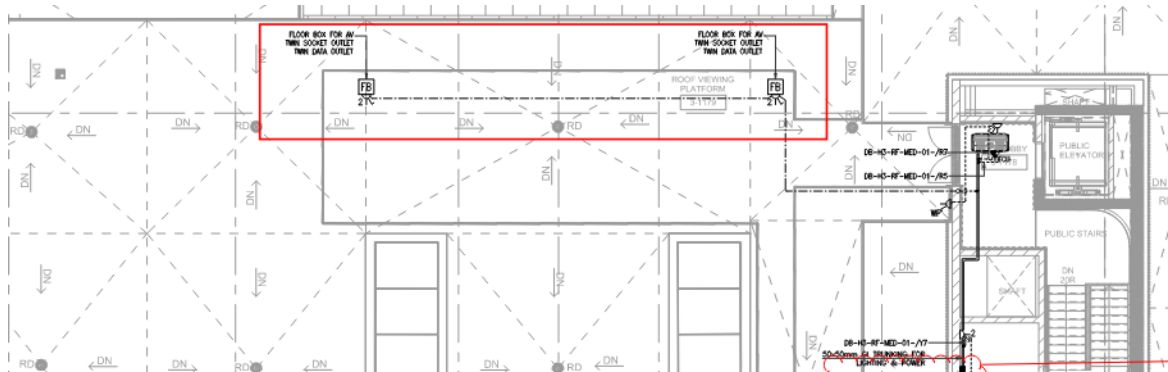
Existing Roof Lighting Control Panel

4.5 ICT / Telecom System

No As-Built Plan specifically the Telecom Schematic Diagram.

The Schematic Diagram is necessary to verify the existing floor level locations of the Telecom Racks or Panels and their existing capacity.

As per below received roof small power layout drawing, there are two floor boxes in roof viewing platform side with twin socket and twin data outlet each for AV/IT. These provisions can be utilized, and additional data points shall be explored once we have the schematic diagram.



Audio- DJ Booths:

Client to provide details of the AV equipment and power requirement can be explored as the design progresses. As per the existing roof DB load schedule, a load of 2kW can be allocated for AV system.

5.0 VT FEASABILITY STUDY

5.1 Key Assumptions

The following key assumptions and benchmarks have been used: -

- Passenger flow into and out of the building from Ground floor only at this stage.
- Population per floor has been based on the defined numbers of persons for the roof top but assumed population level at the Mez-Dining area.
- The population factor has been based on 90% occupation factor as suggested within Cibse Guide D
- Floor to floor heights is not known at this stage therefore SSL have been taken from drawings for this initial analysis only.
- It is assumed that no definitive passenger traffic will come through levels below the main Ground floor level – to be advised
- This analysis has been conducted using just one passenger lift but also, included for the use of the Service lift should the need arise.
- The operating speed for the lift(s) has been analysed between 1.0m/s and 1.6m/s as suggested within BS 5655 pt6 for the height of these buildings.
- Traffic Analysis with a Conventional Lift Control has been used on this building
- Due to the unavailability of lift O&Ms, plus during the site inspection it is noted, there is no manufacture details, model number, occupancy or weight capacity available. As such, we've performed an analysis of the lift shaft & lift car dimensions, the existing lift shaft is suitable for a standard 21 or 24 person lift installed, from the standard Machine Room Less range. This can be further verified through the lift O&Ms.

ID	Level	Height	X/HR	X/HR
Roof	+	19240		
Mez Dining	+	15800	3440	
Ground	+	11000	8240	
Lower Ground	+	6300	4700	
Basement	+	2500	3800	
			P	P

X = Levels served
HR = Headroom
P = Lift Pit

Population 1	Population 2
200	200
100	50
Not Required	Not Required
?	?
?	?

Passenger	Service
-----------	---------

5.2 Traffic Analysis – 250 Persons

Below are the details and Traffic Analysis results of the defined population numbers of 200 persons at the roof bar and an assumed population level of Maximum 50 persons in the Mez-dining area: -

TRAFFIC ANALYSIS RESULTS - 250 Persons							
General Information				Up Peak	SIMULATION RESULTS		
No. of Lifts	Speed (m/s)	Elevator Capacity (kg)	Number of Persons	Handling Capacity (%/5min)	Average Waiting Time (s)	Average Transit Time (s)	Average Destination Time (s)
1	1	1600	21	21.4	24.8	33.7	58.5
	1.6	1600	21	22.6	24.8	33.7	58.5
1	1	1800	24	22.7	21.4	29.6	37.7
	1.6	1800	24	23.9	21.4	29.6	37.7
2	1	1600	21	42.8	9.8	27.9	37.7
	1.6	1600	21	45.1	9.8	27.9	37.7
2	1	1800	24	45.4	8	24.4	32.4
	1.6	1800	24	47.8	8	24.4	32.4

CIBSE Guide D - Preferred Levels	
Interval	60 Seconds or Less
Handling Capacity	6% or Greater
Average Waiting Time	50 seconds or Less
	Compliant
	Close to Compliant
	Fixed Value
	Not Compliant

It can be seen from the analysis results that the single passenger lift traveling at 1.0 or 1.6m/s will provide the required level of service approximately 250 and will 'fully' comply with CIBSE Guide D recommendations to efficiently transport the estimated population through the building and provide a level of Service classed as 'Excellent'.

If the client wants to increase the Level of Service the use of the service lift in conjunction with the passenger lift will be required but then the Level of Service should increase to Better-than-Excellent.

5.3 Traffic Analysis Results – 300 Persons

Below are the details and Traffic Analysis results if the defined population for the roof bar were increased to 250 persons at the roof bar and an assumed population level of 50 persons in the Mez-dining area: -

Traffic Analysis Results

TRAFFIC ANALYSIS RESULTS - 300 Persons							
General Information				Up Peak	SIMULATION RESULTS		
No. of Lifts	Speed (m/s)	Elevator Capacity (kg)	Number of Persons	Handling Capacity (%/5min)	Average Waiting Time (s)	Average Transit Time (s)	Average Destination Time (s)
1	1	1600	21	17.8	29.8	36.1	65.9
	1.6	1600	21	18.8	29.8	36.1	65.9
1	1	1800	24	26.1	26.1	32.7	58.8
	1.6	1800	24	19.9	26.1	32.7	58.8
2	1	1600	21	35.6	11.3	28.5	39.8
	1.6	1600	21	35.6	11.3	28.5	39.8
2	1	1800	24	37.8	9.6	25.6	35.2
	1.6	1800	24	39.7	9.6	25.6	35.2

CIBSE Guide D - Preferred Levels	
Interval	60 Seconds or Less
Handling Capacity	6% or Greater
Average Waiting Time	50 seconds or Less
	Compliant
	Close to Compliant
	Fixed Value
	Not Compliant

It can be seen from the analysis results that the single passenger lift traveling at 1.0 or 1.6m/s will provide the required level of service approximately 300 and will 'fully' comply with CIBSE Guide D recommendations to efficiently transport the estimated population through the building and provide a level of Service classed as 'Excellent'.

If the client wants to increase the Level of Service the use of the service lift in conjunction with the passenger lift will be required but then the Level of Service should increase to Better-than-Excellent.

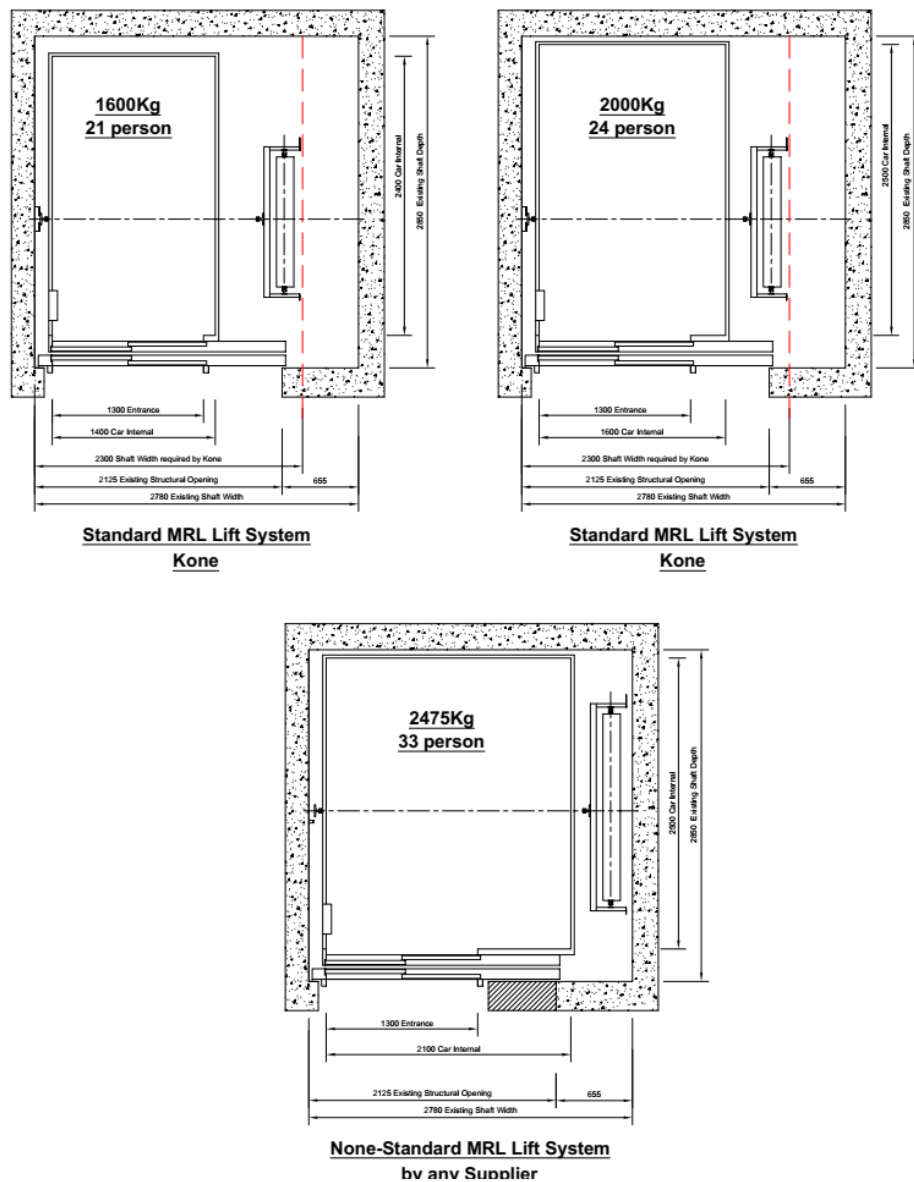
5.4 Lift Shaft Dimensions

As per section 2, Assumptions, item i)

- i) Due to the unavailability of lift O&Ms, plus during the site inspection it is noted, there is no manufacture details, model number, occupancy or weight capacity available. As such, we've performed an analysis of the lift shaft & lift car dimensions, the existing lift shaft is suitable for a standard 21 or 24 person lift installed, from the standard Machine Room Less range. This can be further verified through the lift O&Ms.

Please see below our analysis of the shaft and car dimensions for completeness.

Kindly note, to provide a better use of the existing lift shaft space there is the potential to increase the lift car dimensions to install a larger lift car but, notably there would be a cost implication, to this option.



5.5 Conclusion

The lift systems currently installed within the Milos Restaurant will provide the building & additional rooftop bar with an excellent level of service as indicated by the Traffic Analysis based on the assumed population levels and levels served.

Though as stated in section 5, Lift Shaft Dimensions, to provide a better use of the existing lift shaft space there is the potential to increase the lift car dimensions to install a larger lift car but, notably there would be a cost implication, to this option.

6.0 FLS FEASIBILITY STUDY

6.1 Means of Egress

6.1.1 Occupant Load Factors

As per the UAE Fire code 2018-chapter 3 section 4, the occupant load factors below are used to calculate the number of persons for the rooftop:

Restaurant= 1.4 m²/person

Bar & Terrace=1.4m²/person

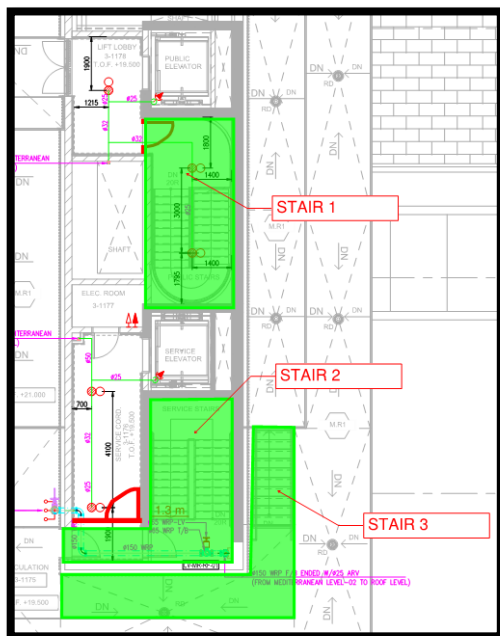
Kitchen= 9.3 m²/person

Egress Capacity Factor:

Stair width per Person: 7.6mm

6.1.2 Means of Egress Analysis for the rooftop

The roof top has access to three staircases as shown below:



The stair exit separation is 16.5m. Therefore, the maximum diagonal distance can be 49.5m.

The total stair width provided is:

Stair 1 – 1.4m

Stair 2 – 1.3m

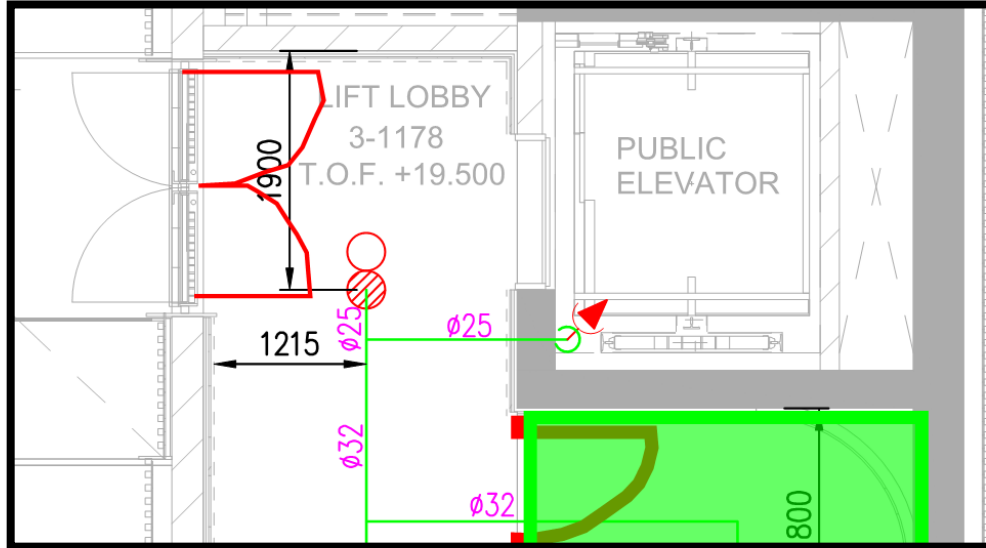
Stair 3 – 1.4m

The total stair width is: 4.1m

The total number of persons that accommodate the roof based on the available stair width is: **539 persons.**

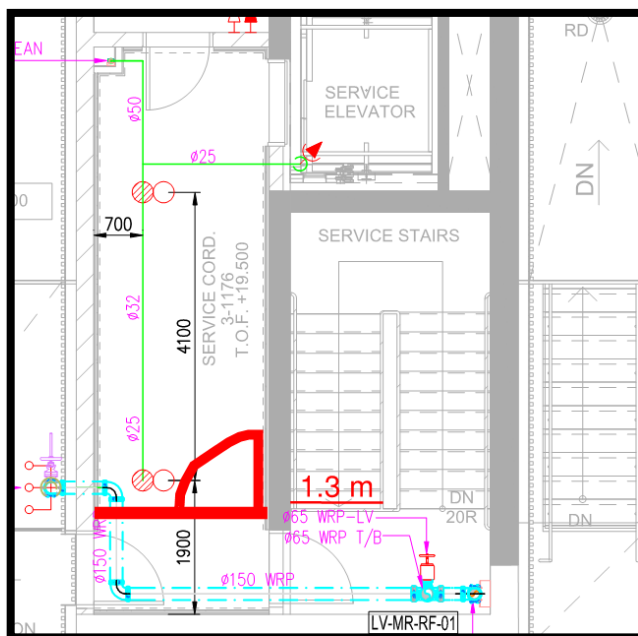
Therefore, 539 persons using an occupant load factor of 1.4m² per person would provide an approximate maximum usable area of 754m².

The door leading to staircase 1 shall be amended to swing in the direction of egress.

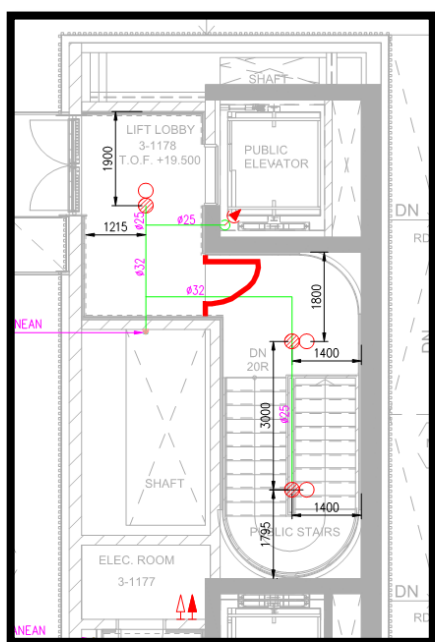


6.2 Fire Rating of walls and doors

In order to access staircase 2, a one-hour fire rated wall and door will be required to be provided, as the room at present has been used for storage, and escaping through storage is not permitted:



Staircase 1 is an open stair which is required to be used as a means of egress (in order to accommodate more occupants), therefore the stair will be required to be provided with a 60-minute fire rated door on the roof level and lower levels.



6.3 Fire Protection requirements

Should there be any enclosed spaces on the roof, a sprinkler system will be required as per the UAE fire and life safety code 2018 edition. As the roof exceeds 50 persons restaurant can accommodate more than 50 persons, it will require to be classified as ordinary Hazard Group 1 and below shall be provided:

- Sprinkler spacing shall be rearrange to 12 m² (4mx3m).
- Ceiling void exceeds 800mm shall be provided with either void sprinkler or compartmentation shall be provided every 280m².
- Kitchen hood suppression shall be provided with the kitchen equipment having 3 or more burners.
- FHC shall cover any part of the building within in 30 meters.
- K type Extinguishers shall be provided in kitchen.

6.4 Fire detection and alarm system requirements

Should there be any enclosed spaces on the roof (dining), a fire alarm system shall be provided as per the UAE Fire code 2018, Chapter 8:

- Fire detection shall be provided thorough the enclosed restaurant space.
- Ceiling void that exceeds 800mm and raised floor exceeds 300mm shall be provided with smoke detectors and Remote Indicators.
- Manual call point shall be provided within 1.5 m from the exits and the travel distances shall not exceed 45m.
- Speaker with flasher shall be provided in the restaurant area.
- All the new devices/ panel shall be connected to the Main fire alarm control panel in the emergency command centre.

7.0 STRUCTURAL FEASIBILITY STUDY

ASG carried out a design analysis model to check the impact of changing the current rooftop converted into a roof top bar/lounge with shade structures.

The below listed As-Built & Shop drawings for the Roof slab, have been used:

As-Built drawings:

As-Built Drawing No.	As-Built Drawing Title	Revision
TRA-SBJ-ABD-CO-B10-02256	Metal Deck Slab at Roof Level (+19.240m)	01
TRA-SBJ-ABD-CO-B10-02257	Metal Deck Slab at Upper Roof Level (+20.740m)	01
TRA-SBJ-ABD-CO-B10-02259	Plan And Sections (+15.800/+15.100m)	01
TRA-SBJ-ABD-CO-B10-02310	Plan View along Grid 942.1 – Sheet (6 of 12)	00
TRA-SBJ-ABD-CO-B10-02320	Column Plan View along Grid 957 – Sheet (1 of 24)	00
TRA-SBJ-ABD-CO-B10-02321	Column Elevation 01 along grid 957 – Sheet (2 of 24)	00
TRA-SBJ-ABD-CO-B10-02322	Column Plan View along Grid 948 – Sheet (3 of 24)	00
TRA-SBJ-ABD-CO-B10-02323	Column Elevation 02 along grid 948 – Sheet (4 of 24)	00
TRA-SBJ-ABD-CO-B10-02324	Column Plan View along Grid 949 – Sheet (5 of 24)	00
TRA-SBJ-ABD-CO-B10-02325	Column Elevation 03 along grid 949 – Sheet (4 of 24)	0
TRA-SBJ-ABD-CO-B10-02340	Column Plan View, Basement to Upp. RF – Sheet (21 of 24)	0
TRA-SBJ-ABD-CO-B10-02341	Column Elevation, Basement to Upp. RF – Sheet (22 of 24)	0
TRA-SBJ-ABD-CO-B10-05725	Level 3 Concrete Plinth – Plans & Sections	0
TRA-SBJ-ABD-CO-B10-02488	Mediterranean Restaurant Basement Level Zone (B10.2) Concrete Foundation Plan	01
TRA-SBJ-ABD-CO-B10-02183	Mediterranean Restaurant Ground Floor Level Zone (B10.2) Concrete Floor Plan	03

Shop drawings:

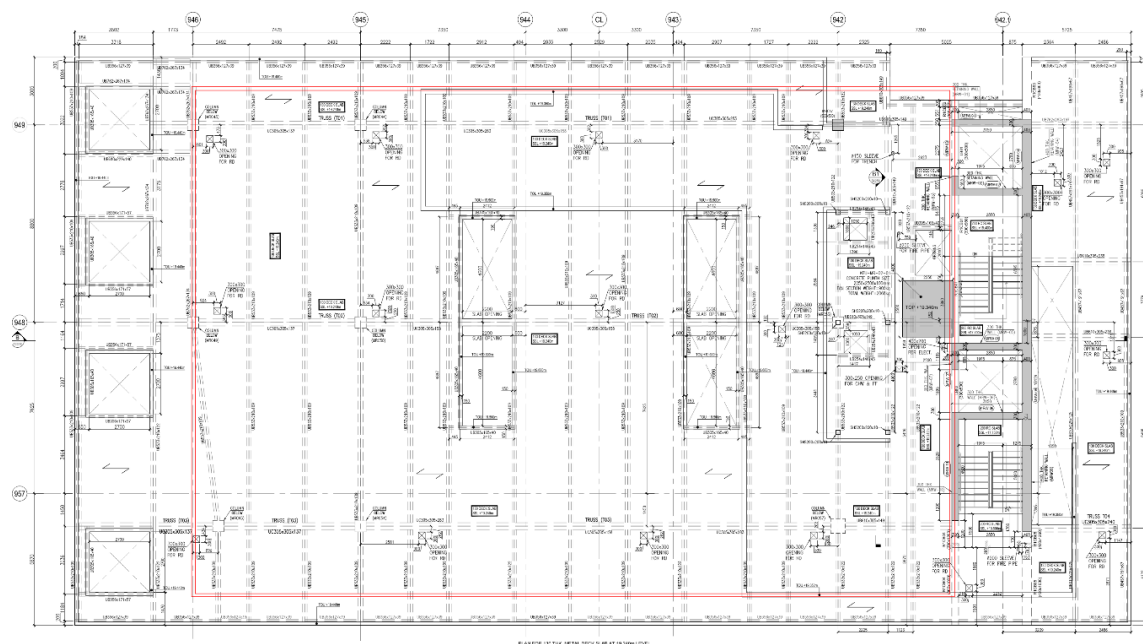
Shop Drawing No.	Shop Drawing Title	Revision
TRA-EVS-SDG-SS-GEN-13571	GA at 2 nd Floor Level - Steel Framing Plan	00
TRA-EVS-SDG-SS-GEN-13572	GA of Roof Truss Bottom Chord Level	00
TRA-EVS-SDG-SS-GEN-13573	GA of Roof Truss Top Chord Level	00
TRA-EVS-SDG-SS-GEN-13574	GA of Upper Roof Level	00
TRA-EVS-SDG-SS-GEN-13575	GA of Truss elevations & Sections	00
TRA-EVS-SDG-SS-GEN-13587	GA of at 2 nd Floor Decking Layout Plan	00
TRA-EVS-SDG-SS-GEN-13588	GA of Roof Truss Top Chord Level	00
TRA-EVS-SDG-SS-GEN-13589	GA of Upper Roof Level Decking Layout Plan	00

The structural system of the existing standalone Milos Greek Restaurant can be summarized as follows:

- Three paralleled Steel Trusses on Grids (948 & 949) and below Grid-957 rested on 8 concrete columns of square shape (0.50 x 0.50 m²) & and one column (0.65 x 0.65 m²).
- The three trusses are supporting Steel beams at unequal spacings varies from 2.4 to 3.3 m.
- 130 mm thick concrete deck slab is used for the entire roof except small part for the staircase landing slab (200 mm Thick).
- Small concrete deck slab at the upper roof level +20.74 at grid-942 rested over 6 No. of steel columns (SHS200x200x10), which are planted over the decking slab steel beams at level+19.24.
- The original design drawings are prepared by (SSANGYONG, BESIX).

7.1 Structural Observation & Assumptions

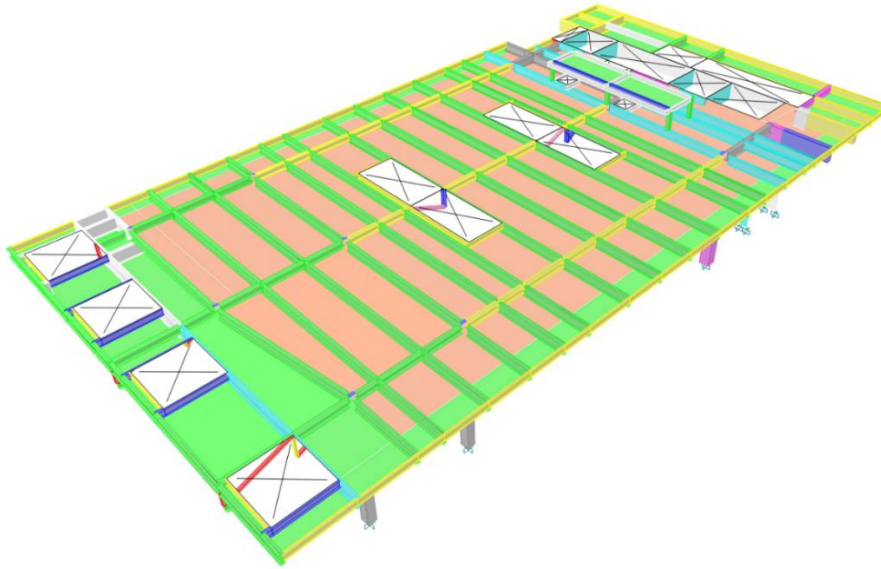
- The steel members grade is not shown on the received As-Built & Shop drawings, ASG has used Grade (S355) as indicated on the received schematic drawings (general notes & legends and abbreviations-2 dwg. No. S002) for the structural analysis & design model.
- ASG considered a live load of 5.0 kN/m² at the new Bar/ Lounge Roof top slab.
- An assumed planter boxes size of (0.6m height x 0.7m width) on the Truss-01 & Truss-03 has been considered.
- The reinforcement of the existing RC columns is not available in the received drawings; a reinforcement ratio of 3.14% for (500x500 columns), and 2.32 % for (650x650) column have been considered in our assessments. However, the actual reinforcement for all columns needs to be verified during the design development stage.
- The existing concrete deck slab of the Roof Top at level +19.24 shall not be used completely for the Milos Greek Restaurant Bar /Lounge, and only a limited area by the red colour indicating new Balustrade on the below figure.



Proposed location for the new Balustrade on existing roof

7.2 Proposed Structural Option 1

A 3D Etabs analysis model has been carried out for Milos Greek restaurant Roof Top to verify the deck slab structural adequacy and present an assessment via the modified use (Bar/ Lounge) that requires higher live load of (5.00 KN/m²). The input & output for the Roof Top slab modeling are summarized as follows:



Etabs Analysis & Design Model – 3D View

Since the base build roof top slab has been designed as un-usable roof and the new modifications required is to change the roof top slab to accessible roof for bar/ lounge use, the deck slab structure have been checked with 5.00 KN/m² live load and some of steel members of the below trusses have been found structurally inadequate.

Design Loads :

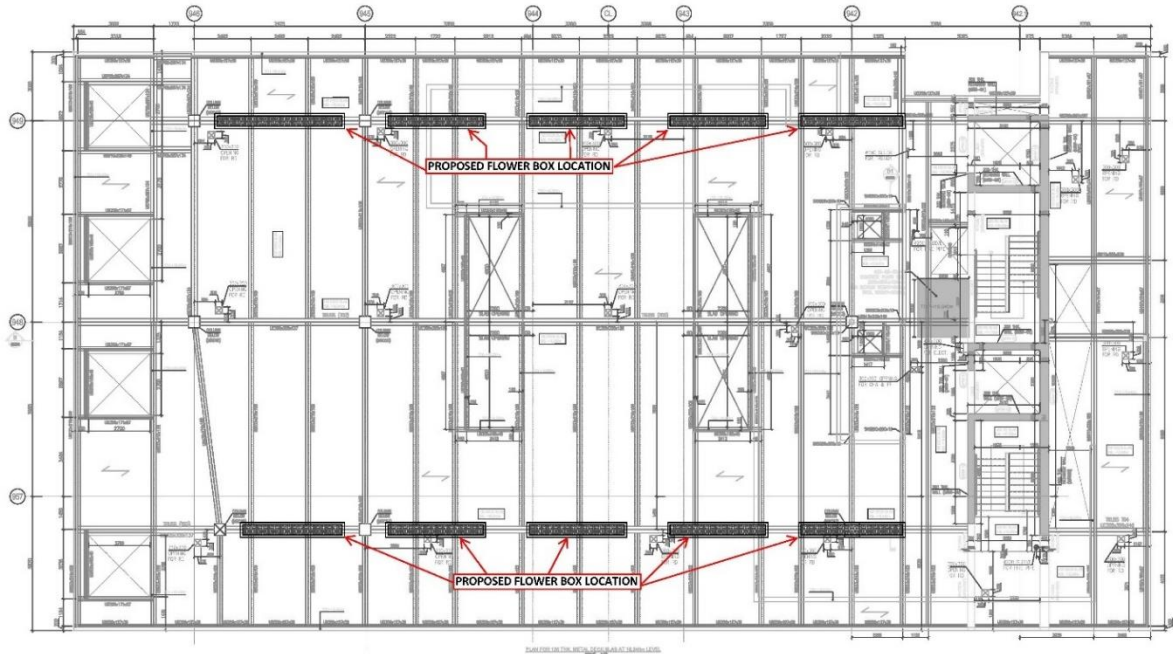
- Super Imposed Dead Loads :
- 1- 130 mm Thk Concrete Deck floorAuto-generated by program
- 2- Roof finishes3.50 KN/m²
- Live Loads 5.00 KN/m²

Roof Top Deck slab @ Level +19.24

Roof Top Deck Slab – Modified Design Live Load

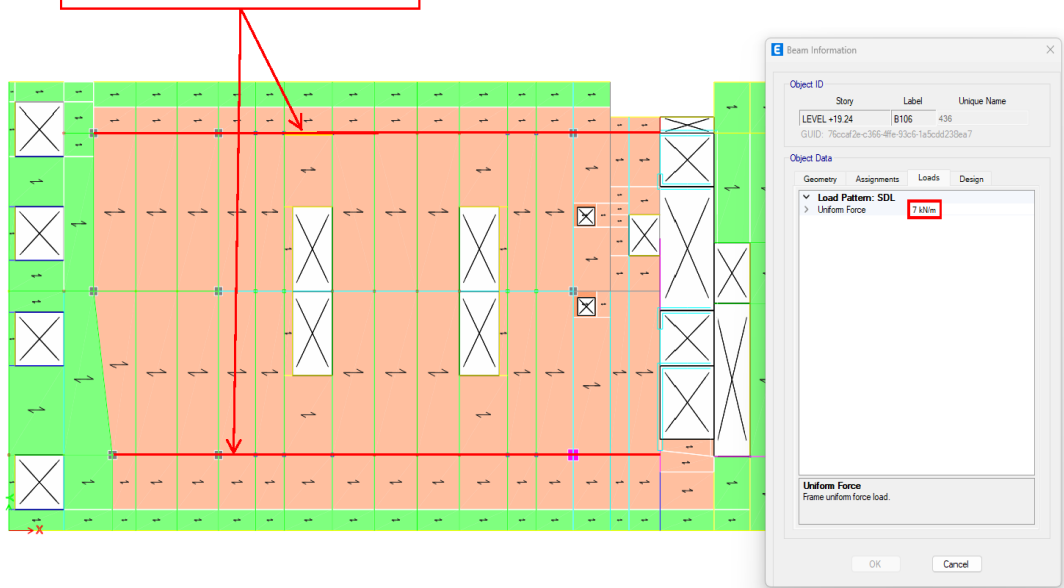
A load for additional planter boxes has been considered as follows:

- The planter box is (0.6m height x 0.7m width), at the proposed location shown in the figure below.
- A line load of 7.0 KN/m' is considered at the proposed locations.



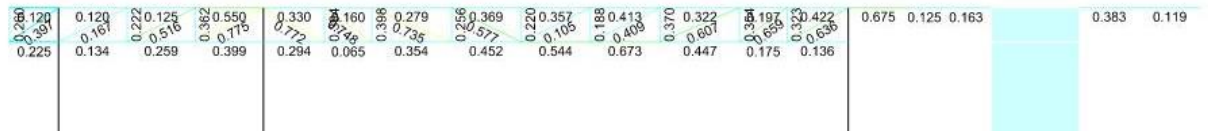
Proposed location for the new Flower Boxes on existing roof

- PROPOSED PLANTER BOXES LOADS :
- Planter Box width = 0.70 m
 - Planter Box Depth = 0.60 m
 - Super Imposed Linear Design Load = 7.00 KN/m'



Roof Top Deck Slab – Additional Planter Box Load

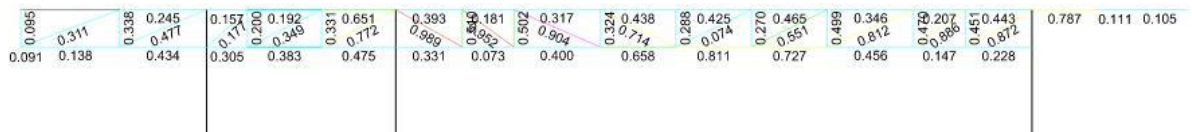
The missing steel members definitions in the As-Built drawings have been assumed in the previous issued Report (Rev.0), and recently such steel members definitions are received in the Shop drawings and used in the updated structural analysis model. Accordingly, some failed members (in the previous) have been found adequate and some other members as shown by red color in the next figures (which are exceeding 1.0) are inadequate and require strengthening or adding new members.



Truss At Grid-949 - Steel Members Design Capacity

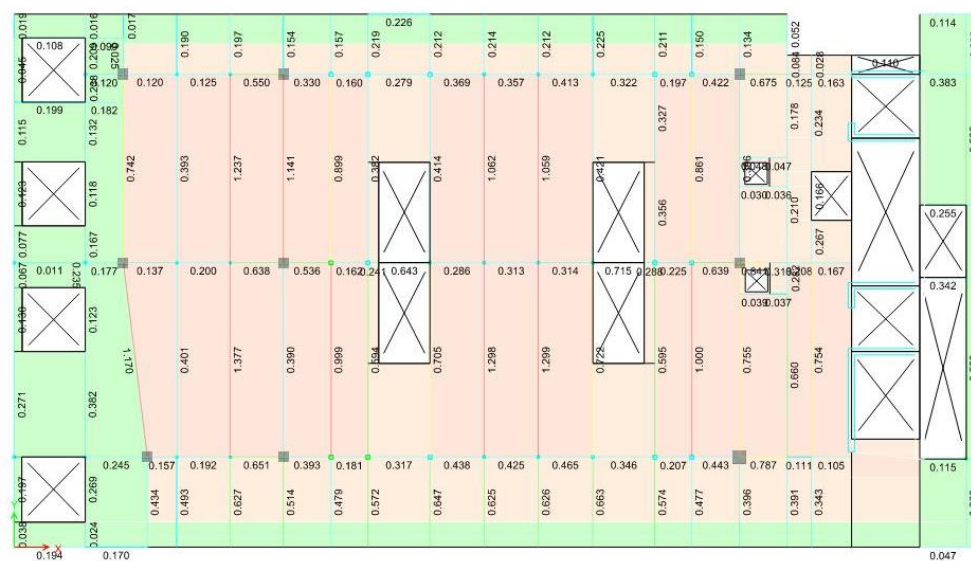


Truss At Grid-948 - Steel Members Design Capacity



Truss At Below Grid-957 - Steel Members Design Capacity

Failed members (At Level +19.24) have been found structurally inadequate as shown by red color in the next figure (which are exceeding 1.0) and require strengthening or adding new members.



Level +19.24 - Steel Members Design Capacity

7.2.1 Structural Conclusion & Recommendations for option 1

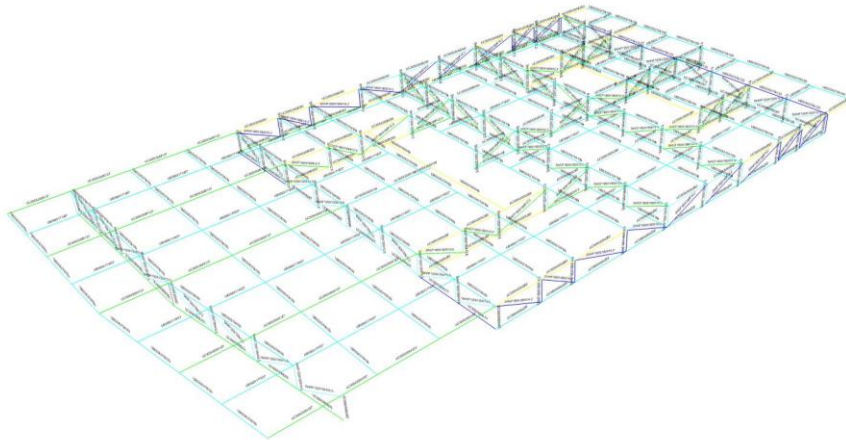
- Some of the steel truss's members are structurally inadequate due to the new design live load for the modified use of the roof top slab and additional planter boxes loads, therefore ASG suggest strengthening work for the steel trusses members to achieve the structural safety.
- The trusses failed members can be strengthened by either adding steel plates or new steel members to achieve the structural safety of entire system.
- The 130 mm thick concrete Deck slab shall need some additional steel secondary beams specifically at the long spacings due to the live load increase.
- The assumed finishing load on the existing roof is 3.5 kN/m², this can be replaced (if possible) to a reduced value of 2.0 kN/m². This has to be checked and confirmed by the design architect and with the local authority regulations.

Therefore, strengthening work will be necessary for some of the existing steel trusses members and addition of new secondary beams to support the existing decking slab to achieve the structural safety. These strengthening works can only be carried out from below, which while structurally feasible this would require removal of large amounts of the existing ceiling and services and as a result would be very disruptive to the existing restaurants operation.

7.3 Proposed Structural Option 2

ASG have studied an alternative option to add a new independent roof deck above the existing roof in order to support the new loads. This would require the new roof to transfer the new loads through to the existing columns below and due to the large spans would need to be based around a structural truss system (similar to the existing steel structure below with some changes).

The depth of the new roof truss is 1.54 m measured from the top steel level to the bottom level of lower chord. It will require a gap of 0.25 m between the bottom chord of the new roof trusses and the existing finishing on site, to avoid any transfer of the loads from the new roof trusses to the existing trusses.



New Roof Top Deck Slab

The design loads have been considered as follows:

- No live load, no finishes load on the existing roof.
- Services/utilities loads only considered on the existing roof.
- Live load of 4.79kN/m² is considered on the new roof.
- Finishes load of 2.5 kN/m² is considered on the new roof.
- The usable area needs to be reduced to be as shown below.

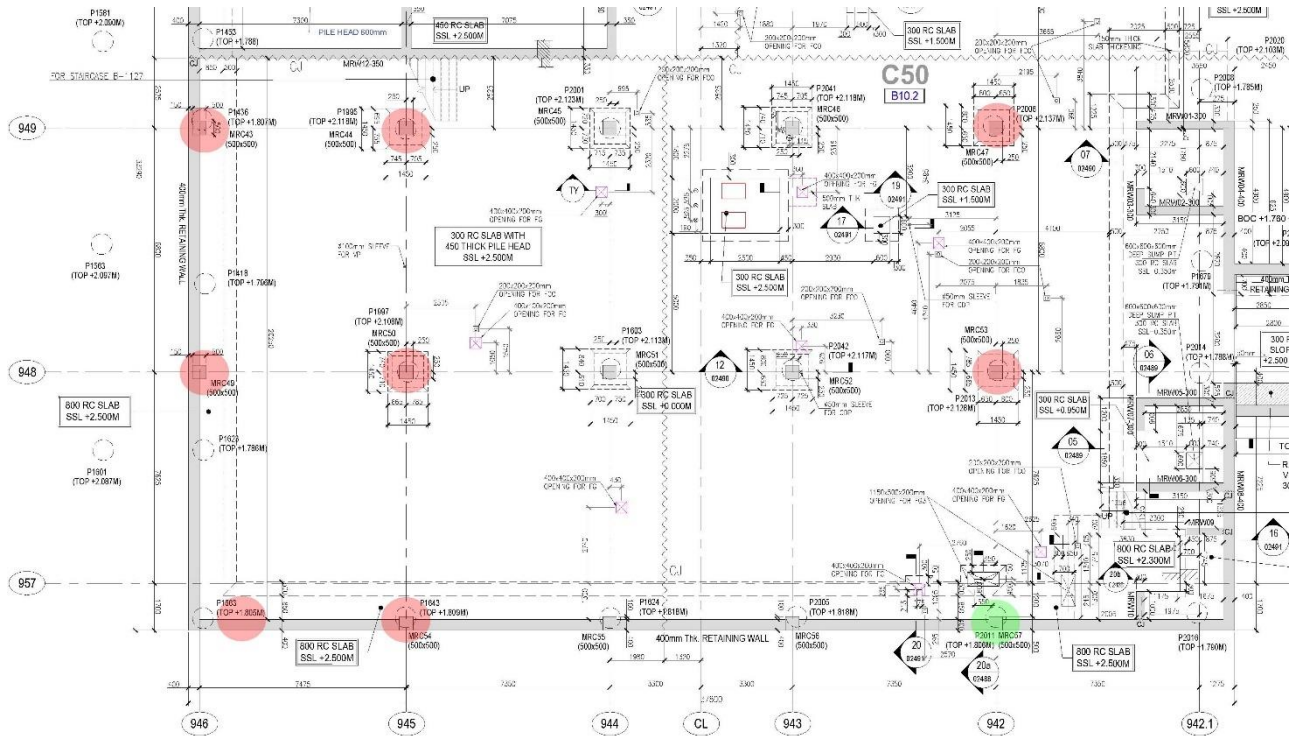


New Trusses & Usable Location at the New Roof

The reinforcement of the existing RC columns has been assumed as:

- 16T25 (3.14%) for the 500x500 mm columns (marked by red below)
- 20T25 (2.32%) for the 650x650 mm columns (marked by green below)

Strengthening for 8 of the existing RC columns (at all floors) is required.



Failed Existing Columns (marked by red)

At the time of writing this report there is no information on the existing piles capacities, we have assessed the loads on the piles as the table below and there is an additional working load $\approx 16\%$ on the existing piles.

Joint At		Reactions (kN)			Total (kN)	% of increase in loads
Horizontal Grid	Vertical Grid	RC Floors	Existing Steel Roof (No Finishes and No live load)	Additional Steel Roof (L.L. =5 kN/m ² & Finishes =3.5 kN/m ²)		
949	946	2467.84	375.8057	259	3052.65	6.1
949	945	2570.90	663.2889	961	4145.19	15.8
949	942	2101.66	625.7599	815	3492.42	16.0
948	946	2510.8294	444.3946	556	3461.22	14.4
948	945	2523.99	525.7643	1124	4123.76	14.0
948	942	2049.19	634.1457	897	3530.33	12.2
957	946	2142.67	564.7597	251	2908.43	6.2
957	945	2491.88	710.9345	946	4098.82	11.5
957	942	2204.16	761.8223	877	3792.99	12.0

7.3.1 Structural Conclusion & Recommendations for Option 2

- All columns of 500x500 mm (8 no. of columns) need strengthening at all floors starting from the foundation (basement floor level).
- For the 8 columns that need to be strengthened This can be carried out by carbon fibre laminate application in order to keep disruption from strengthening to the minimum . ASG will provide the CFRP specialist with all required inputs & outputs to help him in preparing all required drawings & details for construction and authority submission.
- No available information for the capacity of the existing piles. However, an additional load of 10% to 15% on the existing piles is likely to be within the load capacity of the existing pile capacities.
- A comparison between both options can be summarized in the few points as listed below:

	Option 1	Option 2
	Adding additional steel members to the underside of existing steel roof (the existing ceiling would need to be removed)	Adding a complete new independent roof deck above the existing roof supported on a new truss system supported on the 9 existing concrete columns
Additional Loads	4 KN/m ² is added on the existing roof and transferred to the columns, in addition to the own weight of the additional steel members (which is not major).	An approximate additional load of 16.5 KN/m ² will be added on each column, due to the additional roof truss, deck slab, floor finishes & live load.
Existing Columns	All columns are structurally adequate, based on the assumed reinforcement	Need strengthening for 8 columns, starting from foundation level
Installation	Installation of the roof strengthening works required would need to be carried out beneath the existing roof and would be very disruptive to the restaurant.	Installation of the new structural deck would be carried out outside the restaurant but columns strengthening would still be required within the restaurant, but with minor disruption subject to careful planning of this work execution.

For option 2 the existing structure is not able to carry the new steel trusses without strengthening for 8 of the existing RC columns at all floors and verify the additional load on the existing piles, this would have some limited disruption within the existing restaurant.

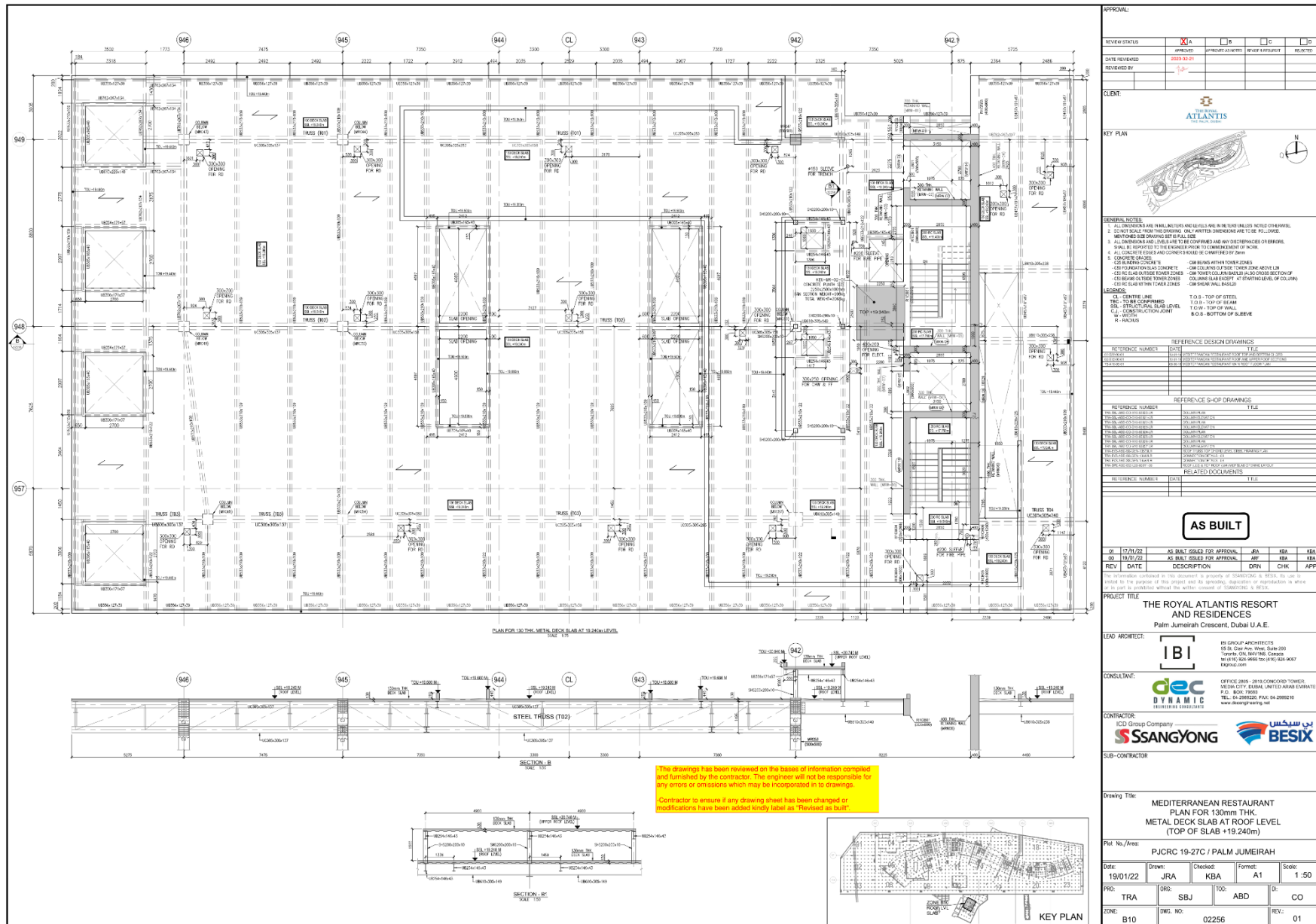
Report Prepared by Astbury Shepherd Gilmour (ASG) Engineering Consultants

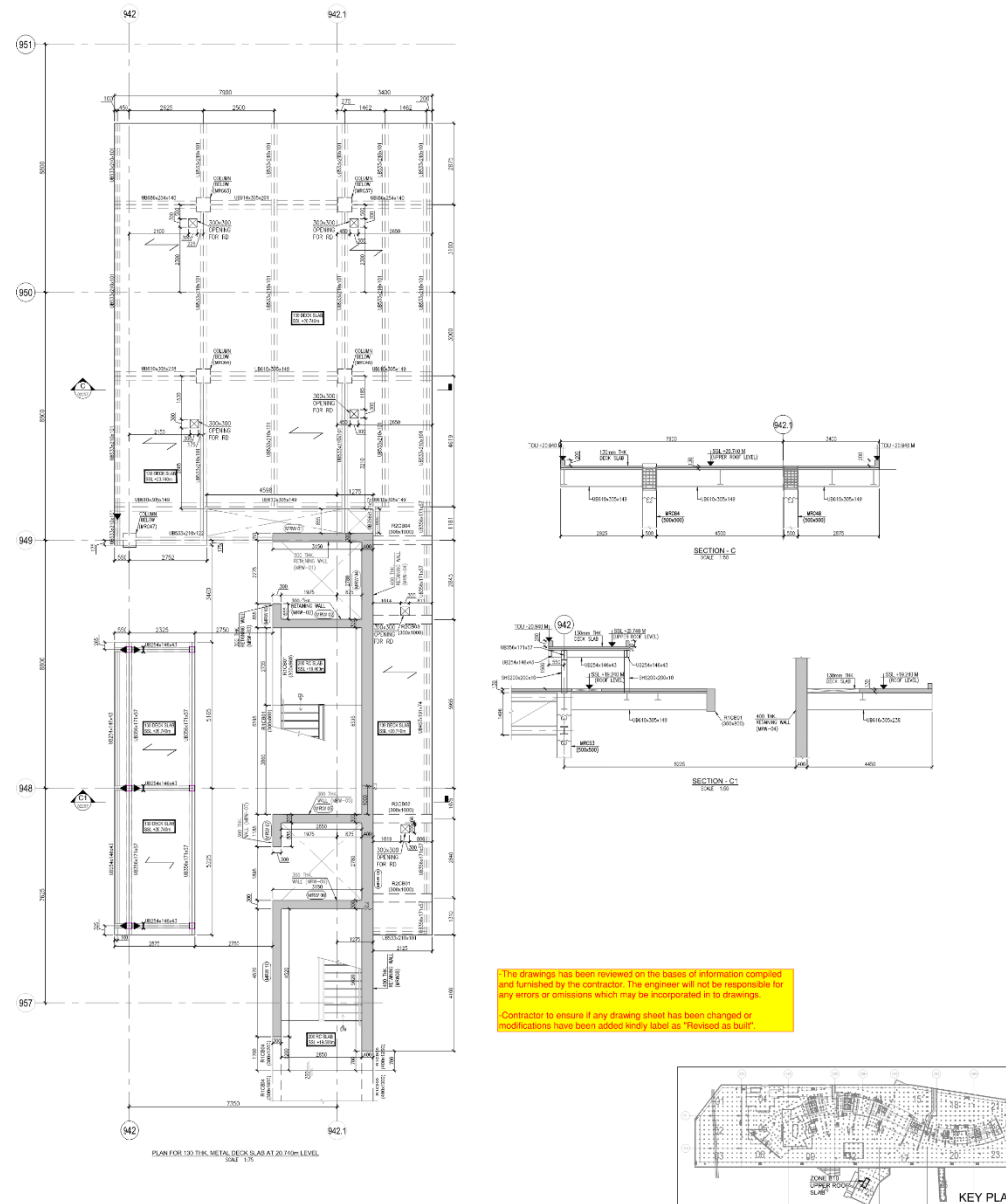
Date: 29th September 2023



APPENDICES

Appendix A – Structural As-Built & Shop Drawings





APPENDIX:				
REVISION STATUS	DATE REVISIONED	REVISION BY	APPROVED BY	REVISIONS
☒ A ☐ B ☐ C ☐ D	APPROVED APPROVED IN ABSENCE REVIEWED REVIEWED IN ABSENCE			
DATE REVISIONED	2023-02-21			
REVISIONED BY	E.S.			
CLIENT:				
 ATLANTIS REAL ESTATE DEVELOPMENT				
KEY PLAN:				
 GENERAL NOTES 1. ALL DIMENSIONS ARE IN MILLIMETERS AND DECIMALS ARE IN METERS UNLESS NOTED OTHERWISE. 2. DIMENT SHALL EXCEED THIS DRAWING, ONLY EXISTING DIMENSIONS AND TOLERANCES SHALL BE MAINTAINED. 3. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. 4. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. 5. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. 6. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. 7. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. 8. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. 9. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. 10. ALL DIMENSIONS SHALL BE MAINTAINED TO THE CENTERLINE OF THE WALL OR CENTERLINE OF THE COLUMN. LEGEND C-1: CENTER LINE C-2: TOP OF SLAB C-3: TOP OF SLAB C-4: TOP OF SLAB C-5: TOP OF SLAB C-6: TOP OF SLAB C-7: TOP OF SLAB C-8: TOP OF SLAB C-9: TOP OF SLAB C-10: TOP OF SLAB				
REFERENCE DESIGN DRAWINGS				
REFERENCE NUMBER	DATE	TITLE	BY	REV
R-001	19/01/22	AS BUILT	E.S.	1
REFERENCE SHOP DRAWINGS				
REFERENCE NUMBER	DATE	TITLE	BY	REV
S-001	19/01/22	AS BUILT	E.S.	1
RELATED DOCUMENTS				
REFERENCE NUMBER	DATE	TITLE	BY	REV
R-001	19/01/22	AS BUILT	E.S.	1

AS BUILT

REV	DATE	DESCRIPTION	DRN	CHK	APP
R	19/01/22	AS BUILT	E.S.	E.S.	E.S.

No information contained in this document is property of Ssangyong & BESIX. No use is permitted in the absence of the written consent of Ssangyong & BESIX.

PROJECT TITLE

THE ROYAL ATLANTIS RESORT AND RESIDENCES

Palm Jumeirah Crescent, Dubai U.A.E.

LEAD ARCHITECT:

IBI GROUP ARCHITECTS
95, Clew Ave, 5th Floor
Dubai, U.A.E.
Tel: +971 4 366 8888 Fax: +971 4 366 8887

CONSULTANT:

DECORATIVE ENGINEERING CONSULTANTS
95, Clew Ave, 5th Floor
Dubai, U.A.E.
Tel: +971 4 366 8888 Fax: +971 4 366 8887

CONTRACTOR:

SSANGYONG CONSTRUCTION CO., LTD.
100, Seongnam-daero, Seongnam-si, Gyeonggi-do, Korea
Tel: +82 31 2000 0000 Fax: +82 31 2000 0001

SUB-CONTRACTOR:

BESIX CONSTRUCTION GROUP
MEDA CITY, DUBAI, UNITED ARAB EMIRATES
Tel: +971 4 366 8888 Fax: +971 4 366 8887

Drawing Title:

MEDITERRANEAN RESTAURANT
PLAN FOR 1500mm THK
METAL DECK SLAB AT UPPER ROOF LEVEL
(TOP OF SLAB +20.740mm)

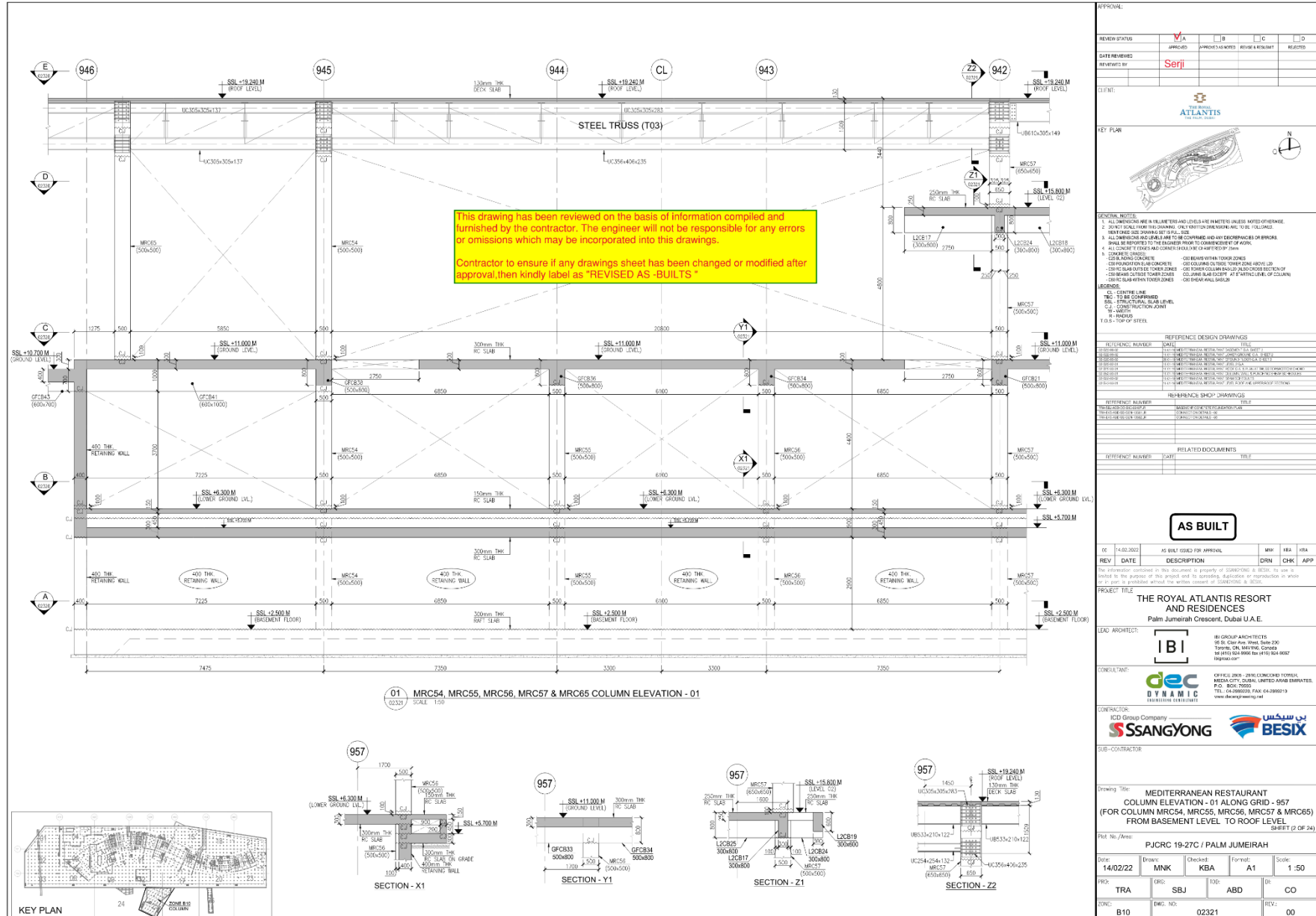
Plot No./Rev:

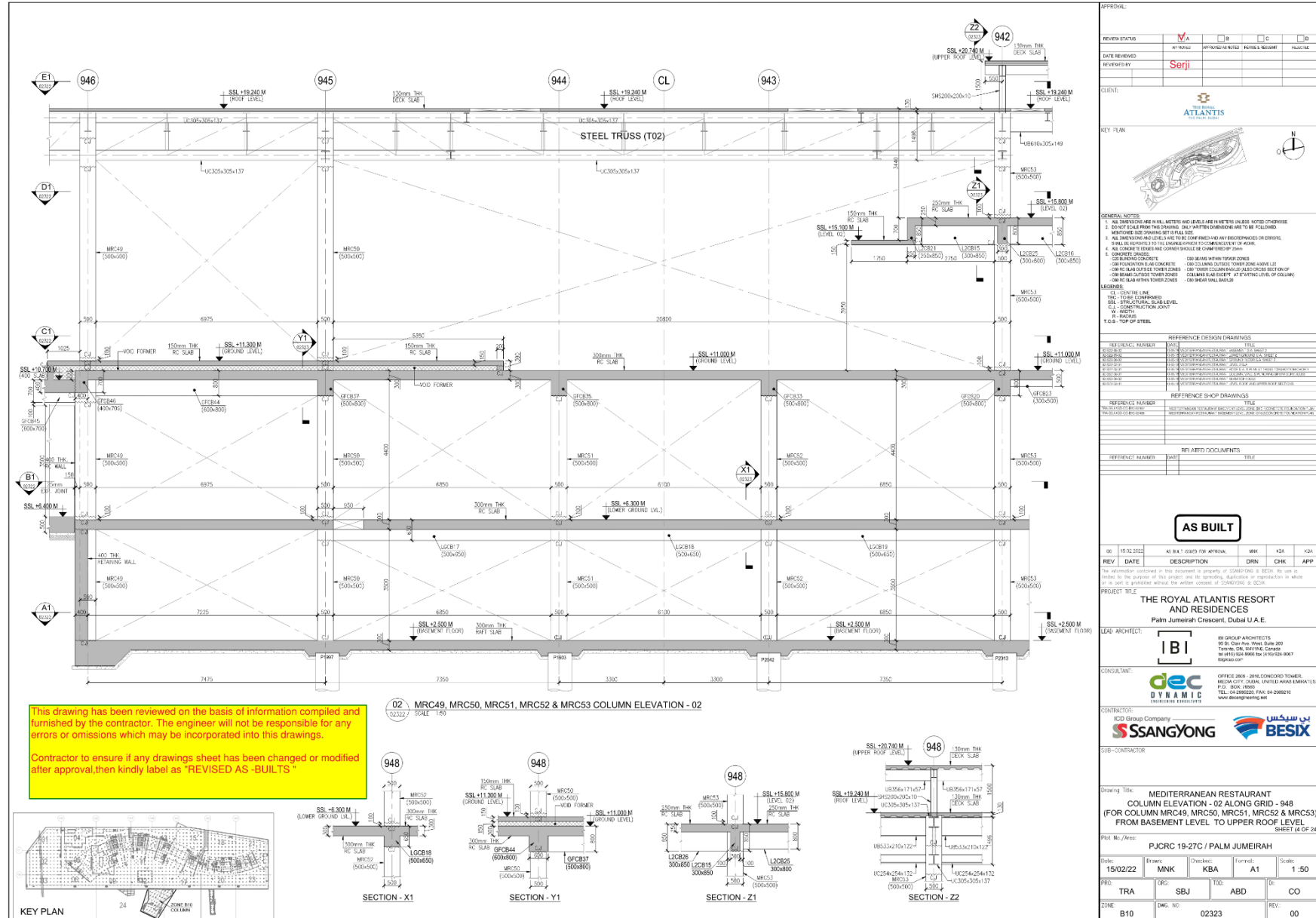
PJCR-19-27C / PALM JUMEIRAH

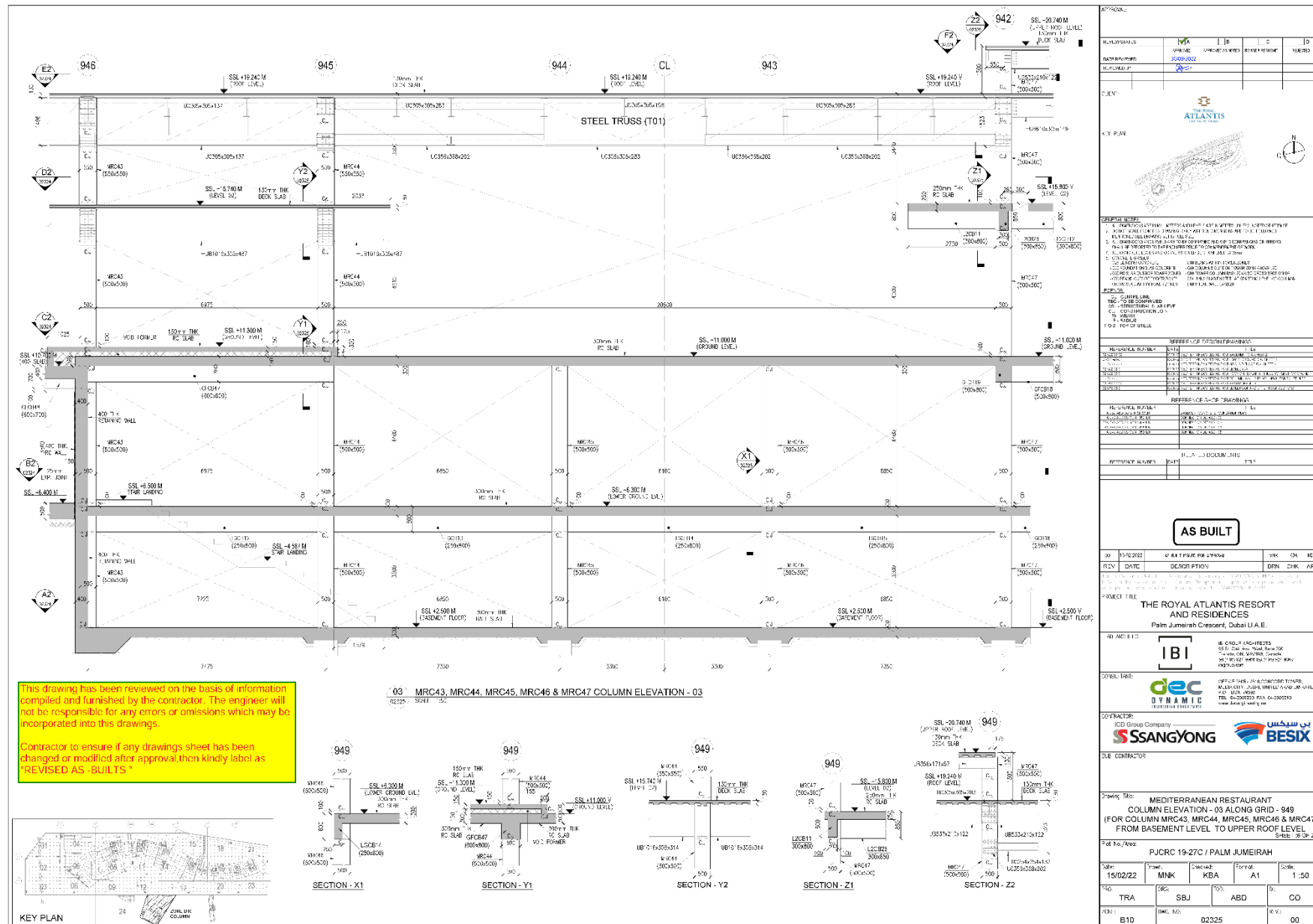
Date:	Drawn:	Checked:	Format:	Scale:
19/01/22	JRC	TBA	A1	1:75

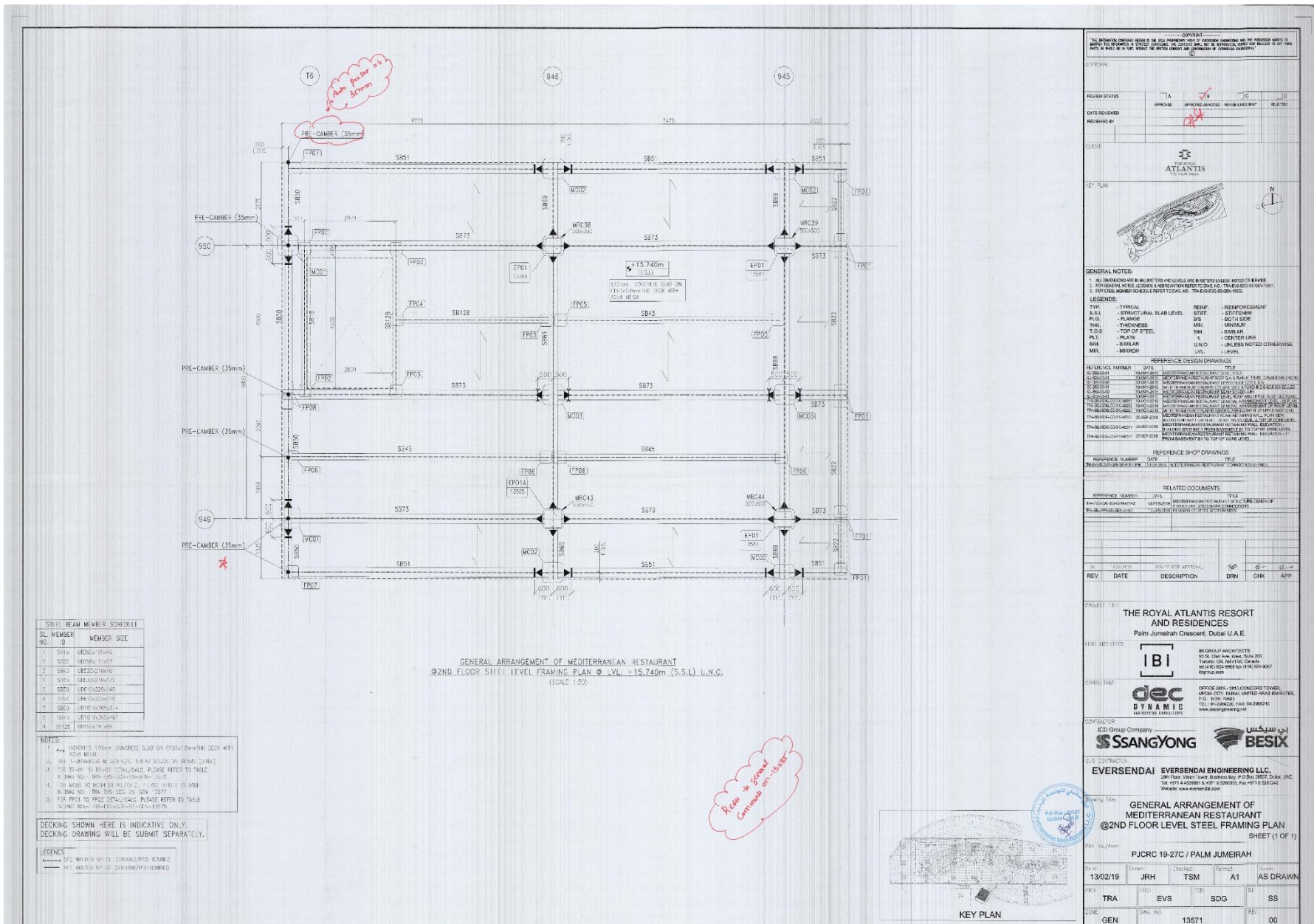
PRO:	DRA:	CHK:	APP:
TRA	SRJ	ABD	CO

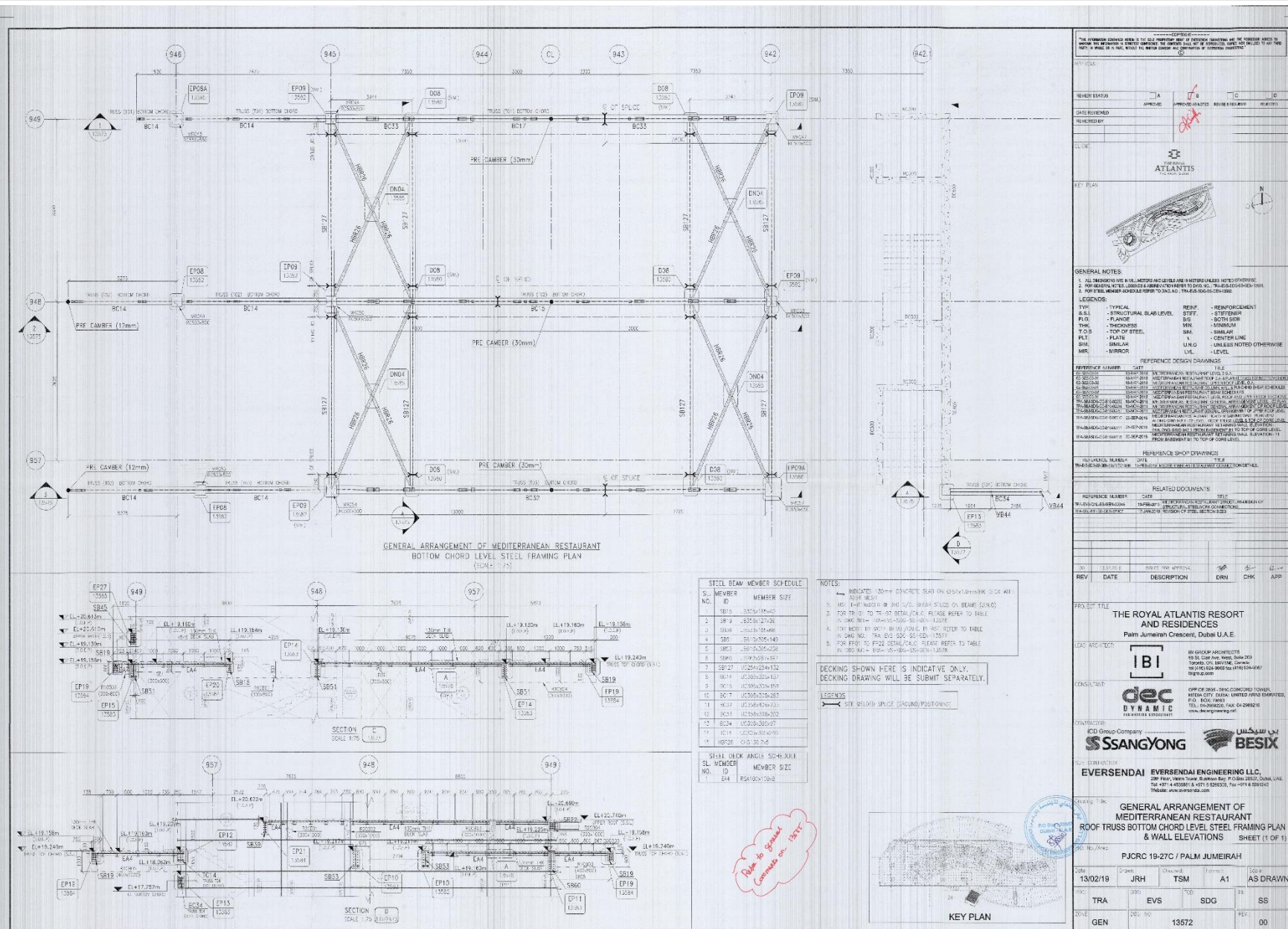
ZONE:	B10	DWS:	02257	REV:	01
B10	DWS:	02257	REV:	01	

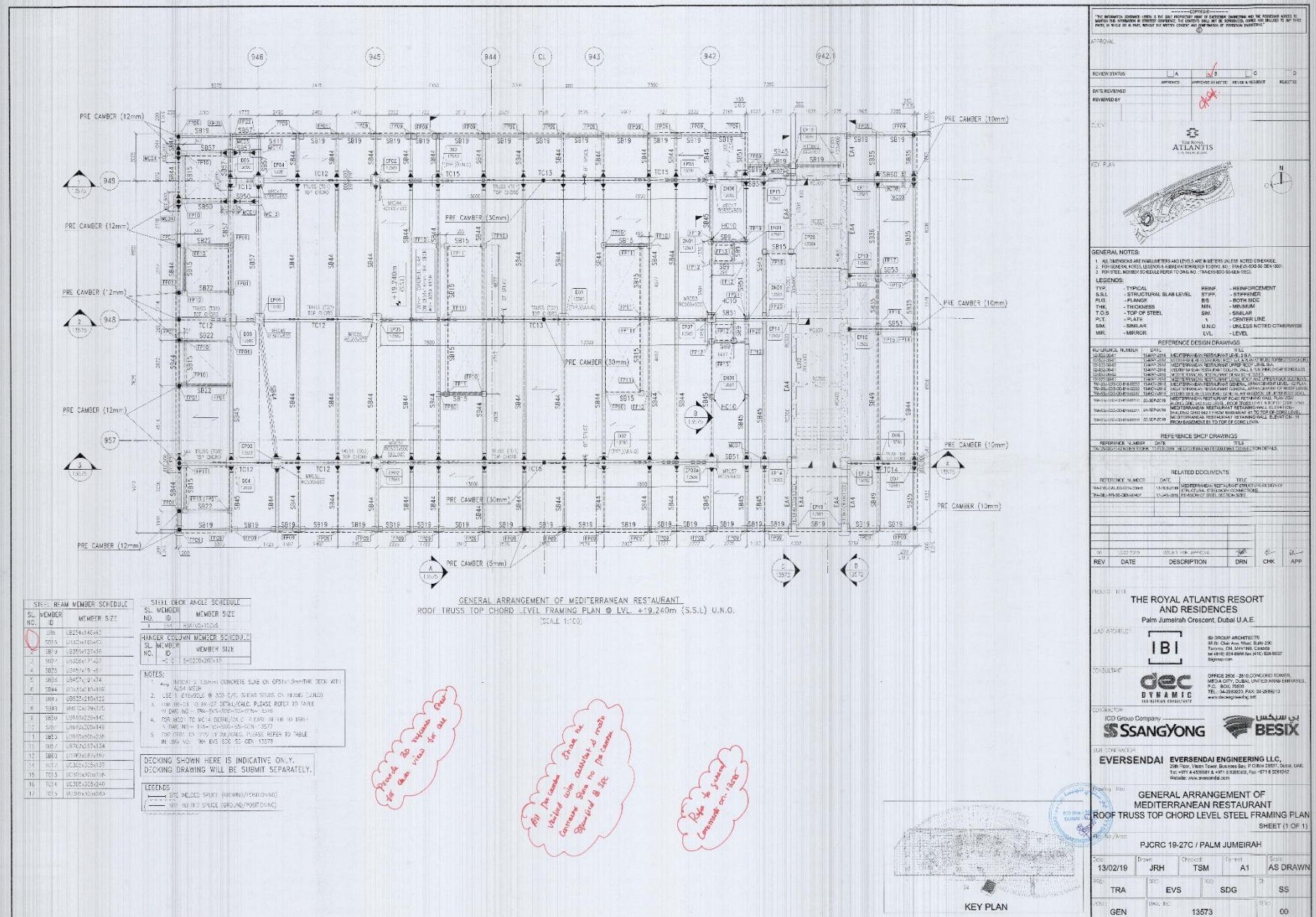


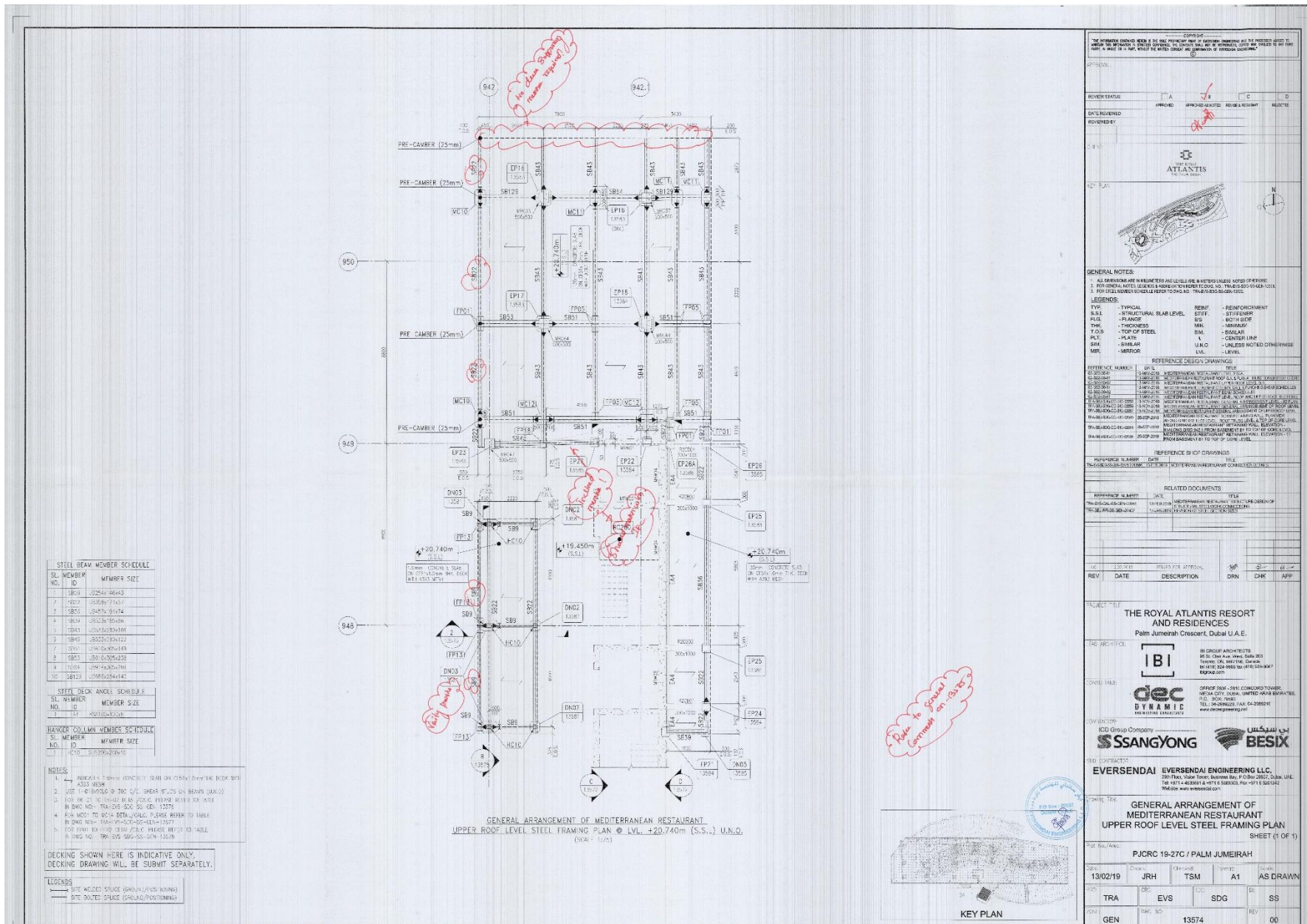


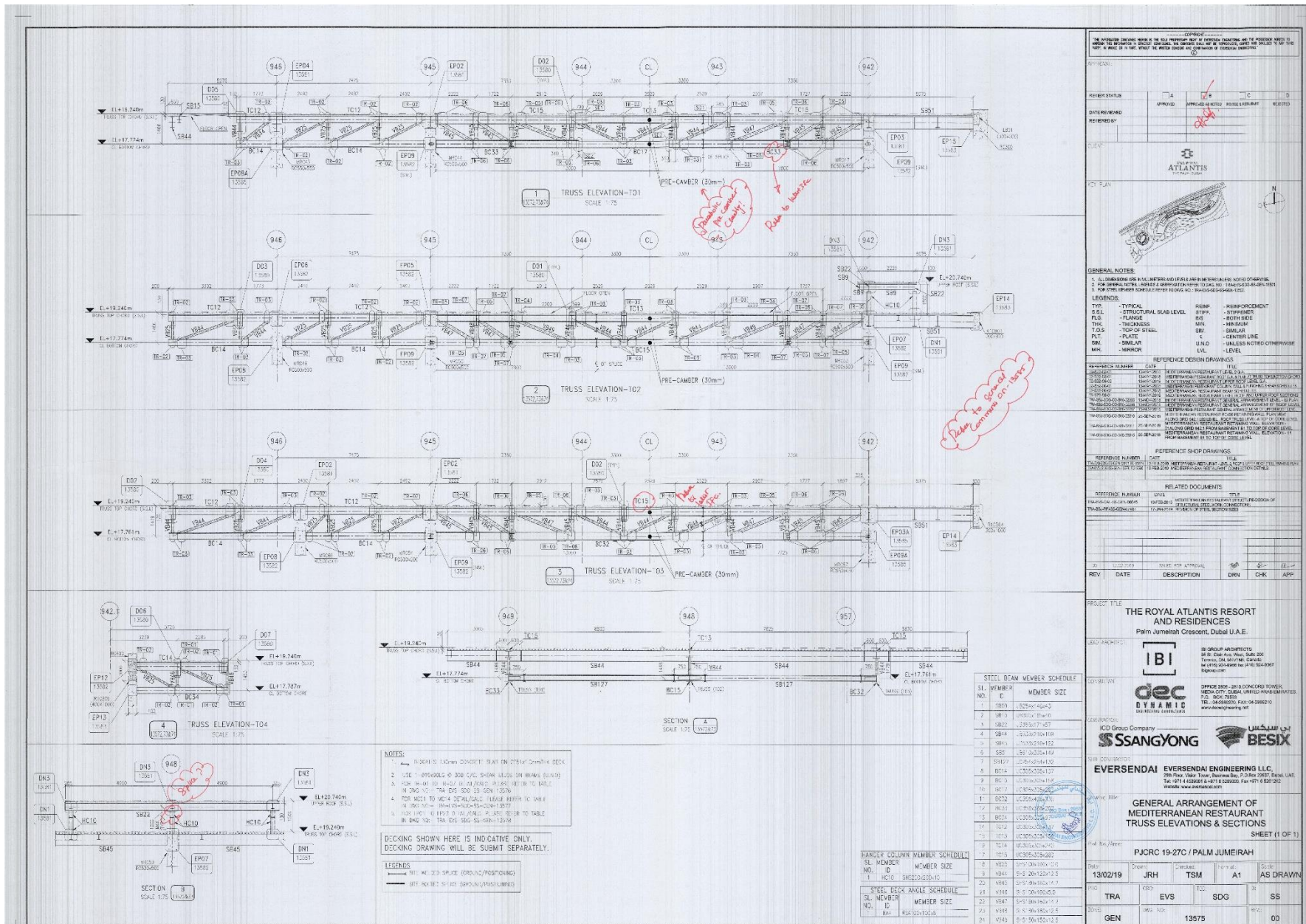


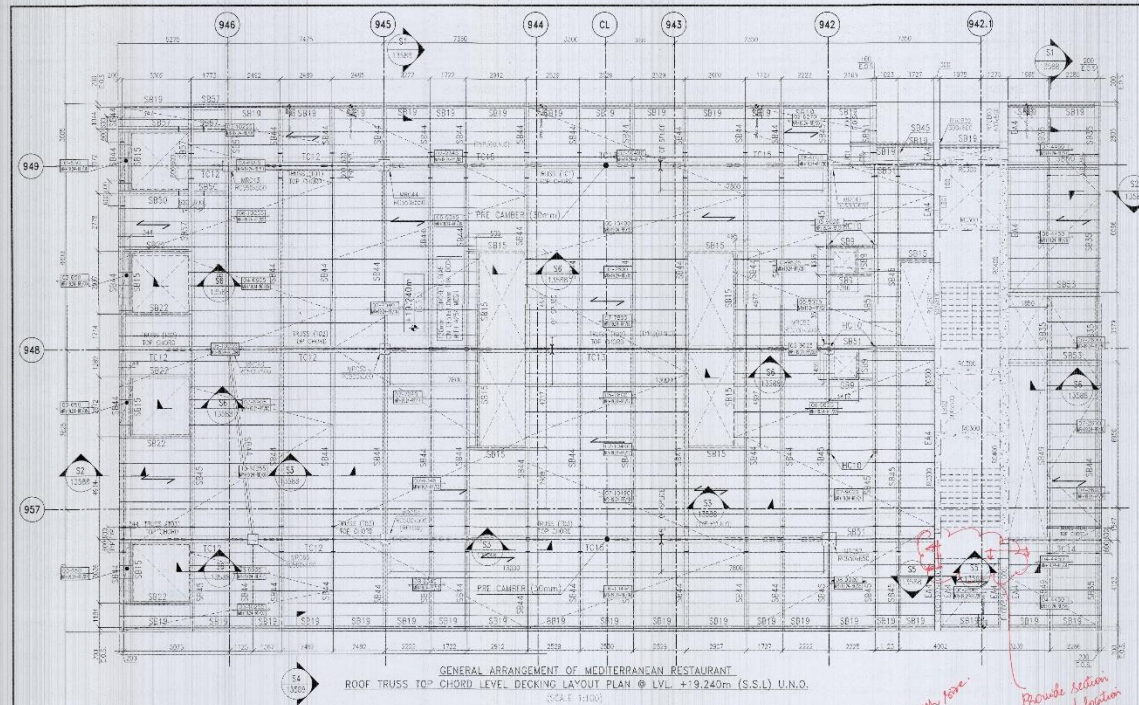






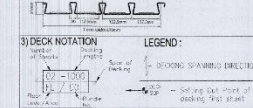






DECKING NOTES:

- 1) ALL DIMENSIONS IN MM AND LEVELS IN METERS U.N.O.
- 2) ConFlor S1 (1.0mm thick) Steel Composite Deck profile



4) FIXING INFORMATION FOR SHALLOW DECKING

FIXING INFORMATION FOR SHALLOW DECKINGS

To steel	Heavy duty powder anchored bolts - 1400 (6.8-16) 10-15% needed (check on equivalent for different lengths) 6.8-16, 10-15% needed through other structural steel to be loaded - 1400
Steel anchorage	Steel anchorage to steel plate - 1000 (6.8-16) 10-15% needed through other structural steel to be loaded - 1400
Majority of anchors	1000 (6.8-16) 10-15% needed through other structural steel to be loaded - 1400
to steel plate	1000 (6.8-16) 10-15% needed through other structural steel to be loaded - 1400

FIXING INFORMATION

Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Decking	Deck
---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	---------	------

